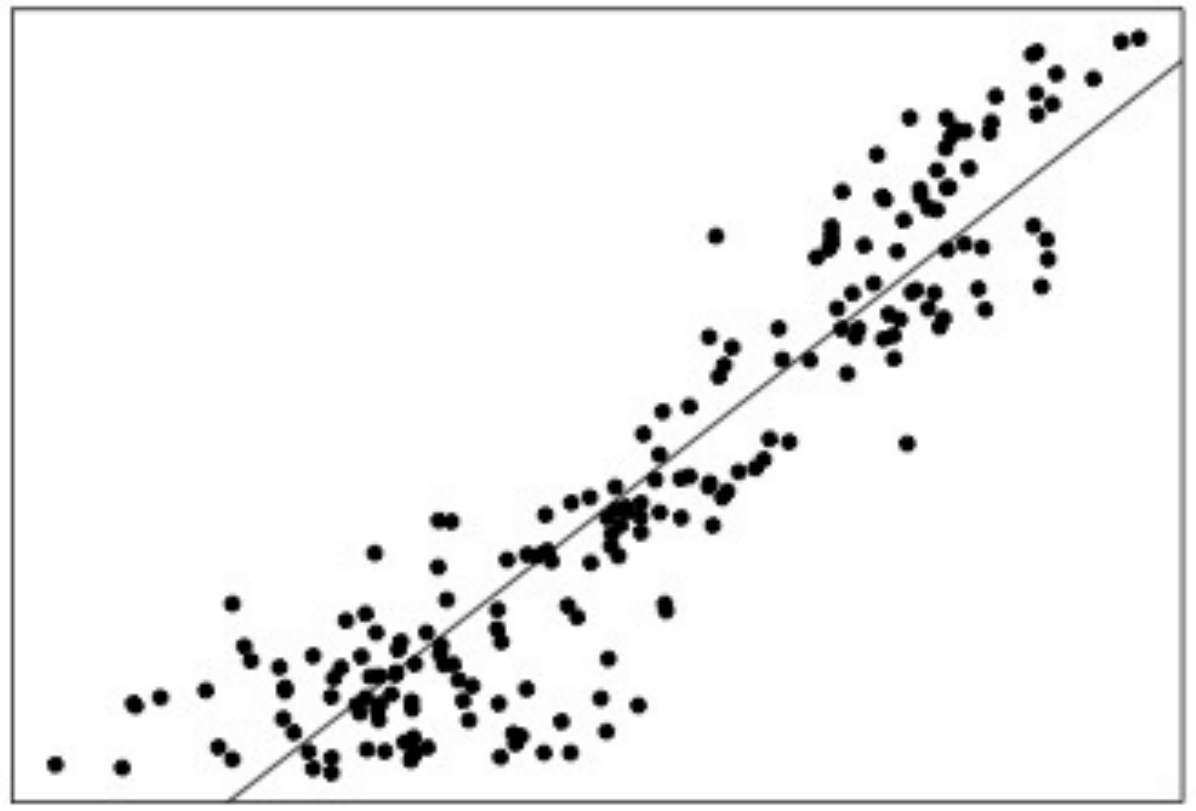
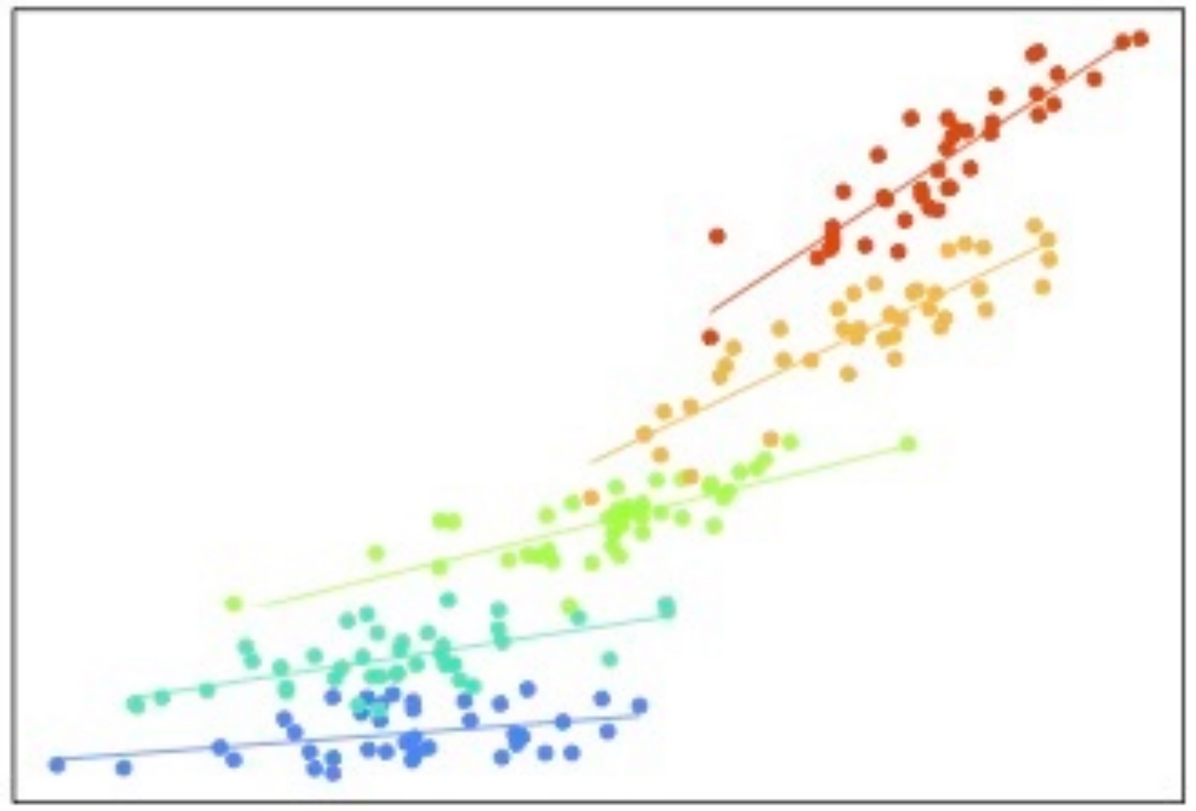
AN INTRO TO MULTILEVEL MODELING

IMPS

July 20, 2026

AGENDA





- Introduction to Multilevel Data
- The Null Model and ICC
- Level 1 and Level 2 Predictors
- Random Slopes
- Estimation
- Model building

Eastern Time

Break: 11:45 - 12:00

Lunch: 1:30 - 2:30

Break: 3:45 - 4:00

Hour

Break: 1h45m - 2h

Lunch: 3h30m - 4h30m

Break: 5h45m - 6h

INTRODUCTION TO MULTILEVEL DATA

SIMPLE RANDOM SAMPLES

In simple random sampling, all units have equal probability of being selected for the sample.

Units are sampled independently.

MULTISTAGE SAMPLES

Multistage sampling selects a sample in more than one step; clusters are sampled, and then the units within the clusters are sampled.

Individual units do not have the same probability of being selected.

Units may not be independent.

MULTILEVEL DATA RESULTS FROM MULTISTAGE SAMPLING

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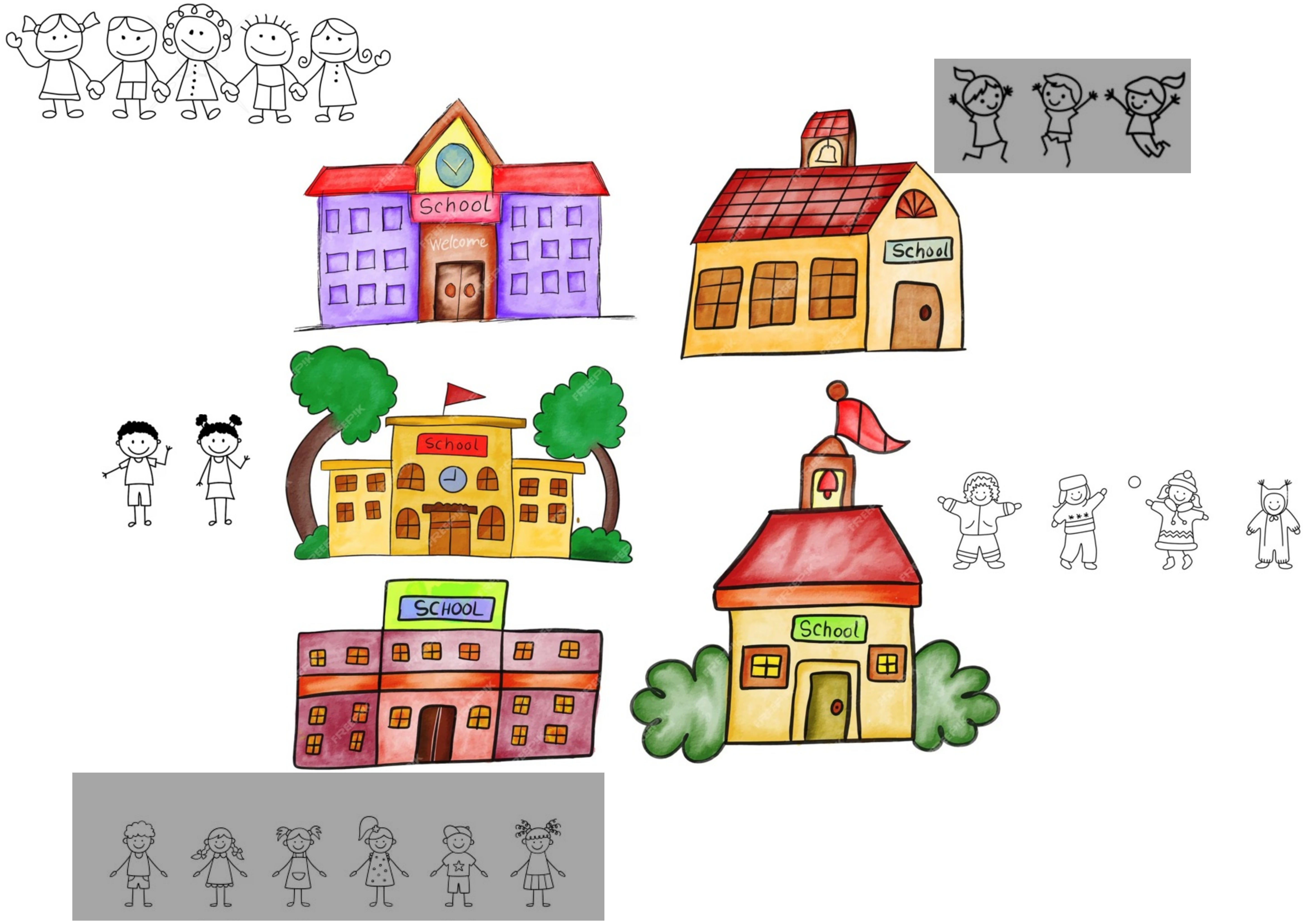


DEPENDENCE AS A NUISANCE

When we have multilevel data, we have observations that are likely more similar to observations in the same cluster.

If we assume our data are independent and use methods that assume independent observations, we are violating assumptions! One consequence is that our standard errors will be too small. We increase our risk of Type I errors.

Therefore we need other methods that can accommodate the data structure that we have!

DEPENDENCE AS A NUISANCE

When we have multilevel data, we have observations that are likely more similar to observations in the same cluster.

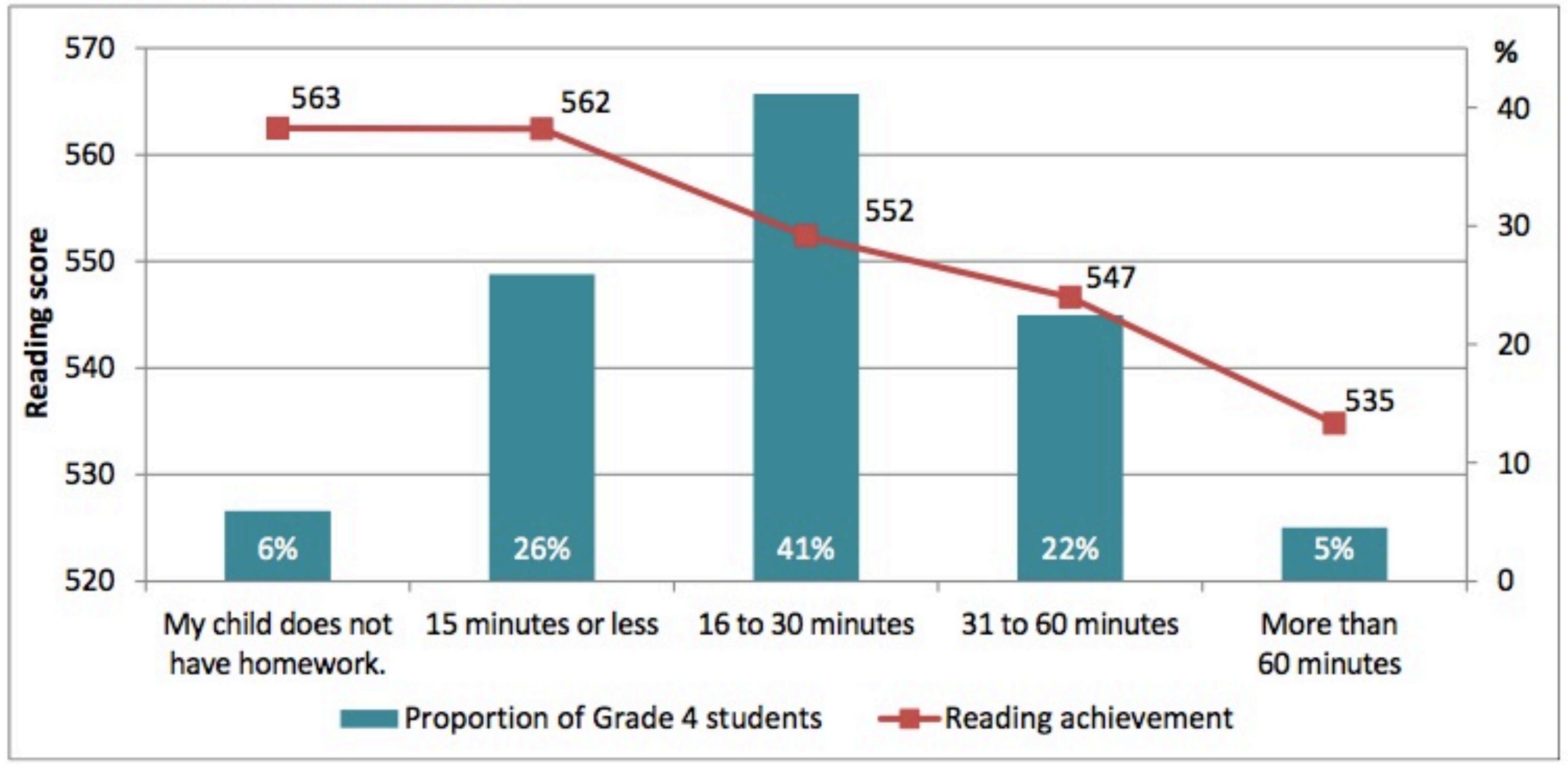
If we assume our data are independent and use methods th… …dent observations, we are violating assumptions! One cons… errors will be too small. We increase our risk of Ty…

Reason #1 for Multilevel Models!

Therefore we need other methods that can acco… have!

AN ASSOCIATION BETWEEN HOMEWORK AND TEST SCORES?

CHART 1 **PIRLS 2011[1] – Reading achievement scores by amount of time spent per day on homework (as reported by parents or primary caregivers)**



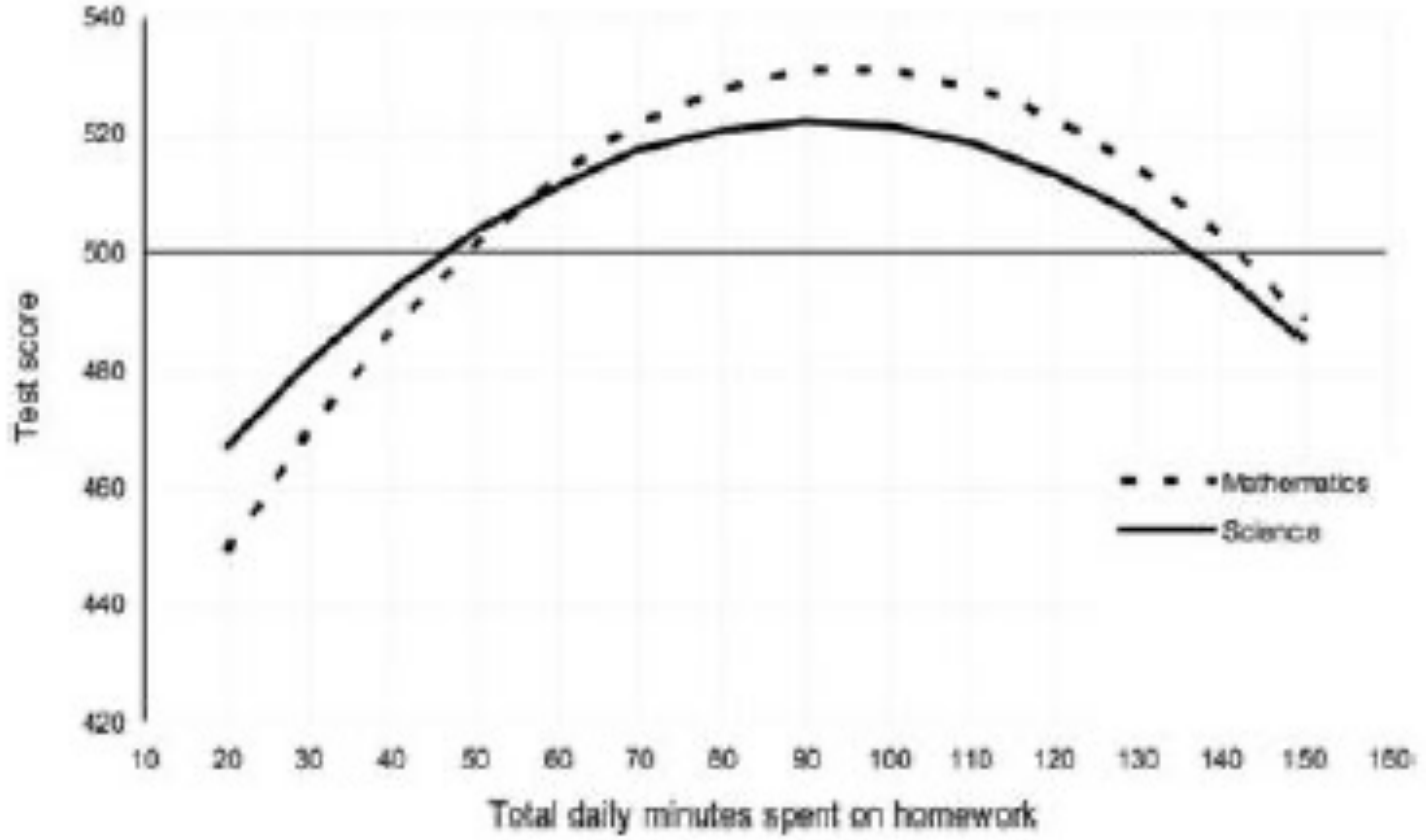
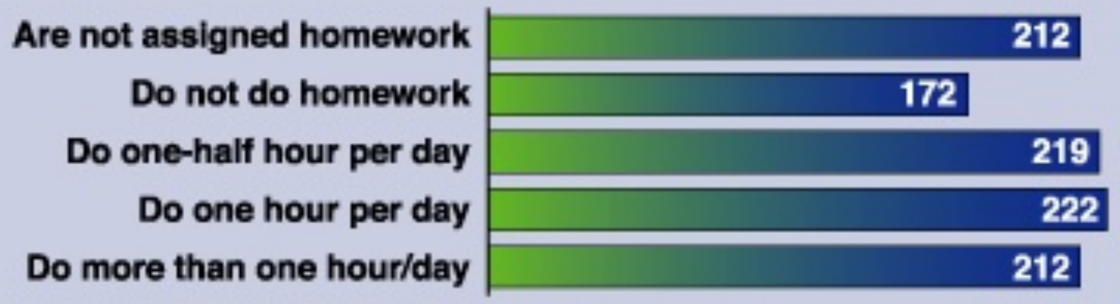


Figure 2. Prediction of performance as a function of total daily time spent on homework.

As Homework Increases, Test Scores Drop

After about an hour, homework becomes counterproductive — at least for fourth-graders, according to an Education Department study. Fourth-graders who spent less than an hour a day on home-work did better on reading tests than students who did more. And students who had no homework scored the same as students who did more than an hour a day.

Reading Test Scores for Various Homework Levels
(for fourth-graders)



Source: National Center for Education Statistics, National Assessment for Educational Progress, "1992-2000 Reading Assessments," August 2000

THE EDUCATIONAL LONGITUDINAL STUDY (2002)

- High school sophomores surveyed and assessed in 2002
- Parents, teachers, principals and librarians also surveyed
- 752 schools and 15,362 students in the sample
- Students were surveyed again in two years later (F1) and transcripts were collected (if students were in the same school)
- Students were surveyed again in two more years later (F2) and post-secondary education institution was collected (if applicable).
- We will be working with a subset of the data from 2002 (BY) and 2004 (F1 for first follow-up.

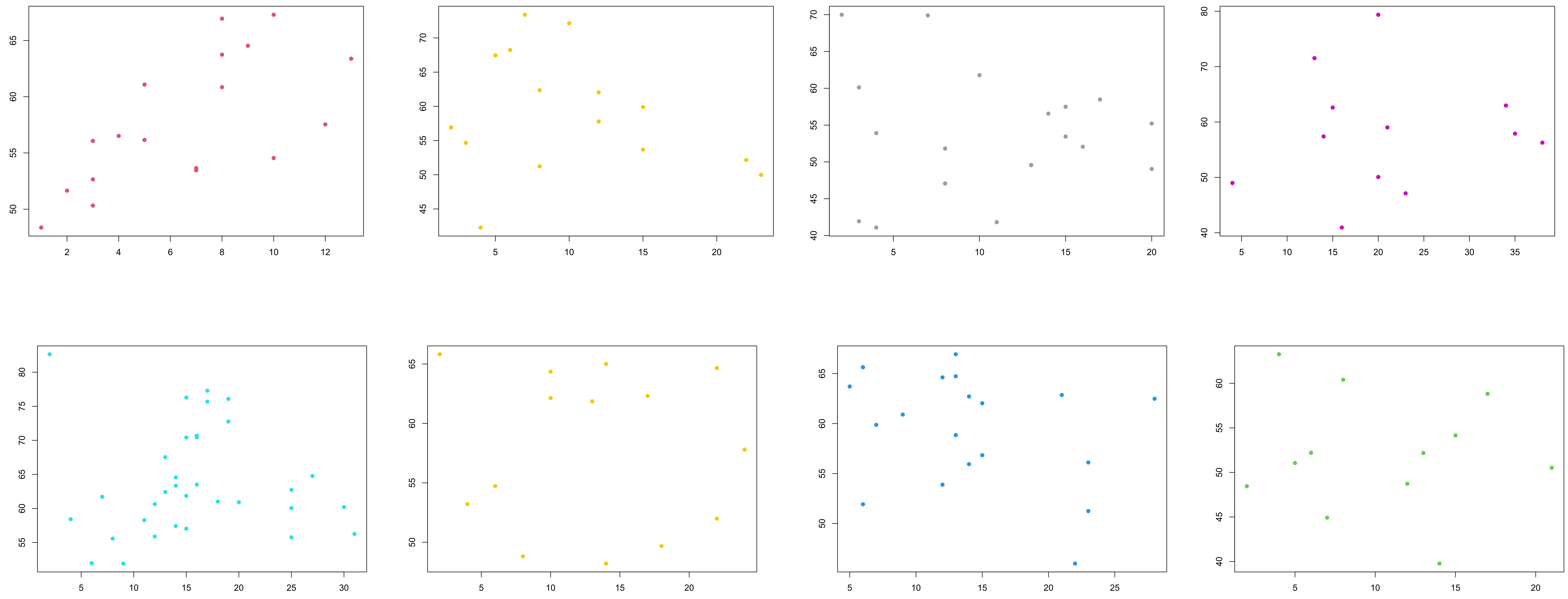
We are IGNORING important complex sampling design issues (weights, stratification, etc.) DO NOT think that these are "real" analyses we are doing.

THE EDUCATIONAL LONGITUDINAL STUDY (2002)

Variable Name	Description
STU_ID	Student id
SCH_ID	School id
male/female	Sex of student indicators
AmIndian/Asian/Black/ Hispanic/Multi/white	Race/ethnicity indicators
pared	Parent education level
income	Income level
riskfc	Number of risk factors
mathSE	Math self efficacy
hmwrk	Number of hours spent doing homework each week
wkrhrs	Number of hour spent working at a job each week
ses	Measure of socio-economic status
math	Math test score
read	Reading test score
f1math	Follow up math test score
f1everdo	Follow up drop out indicator
f1mathse	Follow up math self efficacy
private	School is private
rural	School is located in a rural area

MODELING INDIVIDUAL SCHOOLS

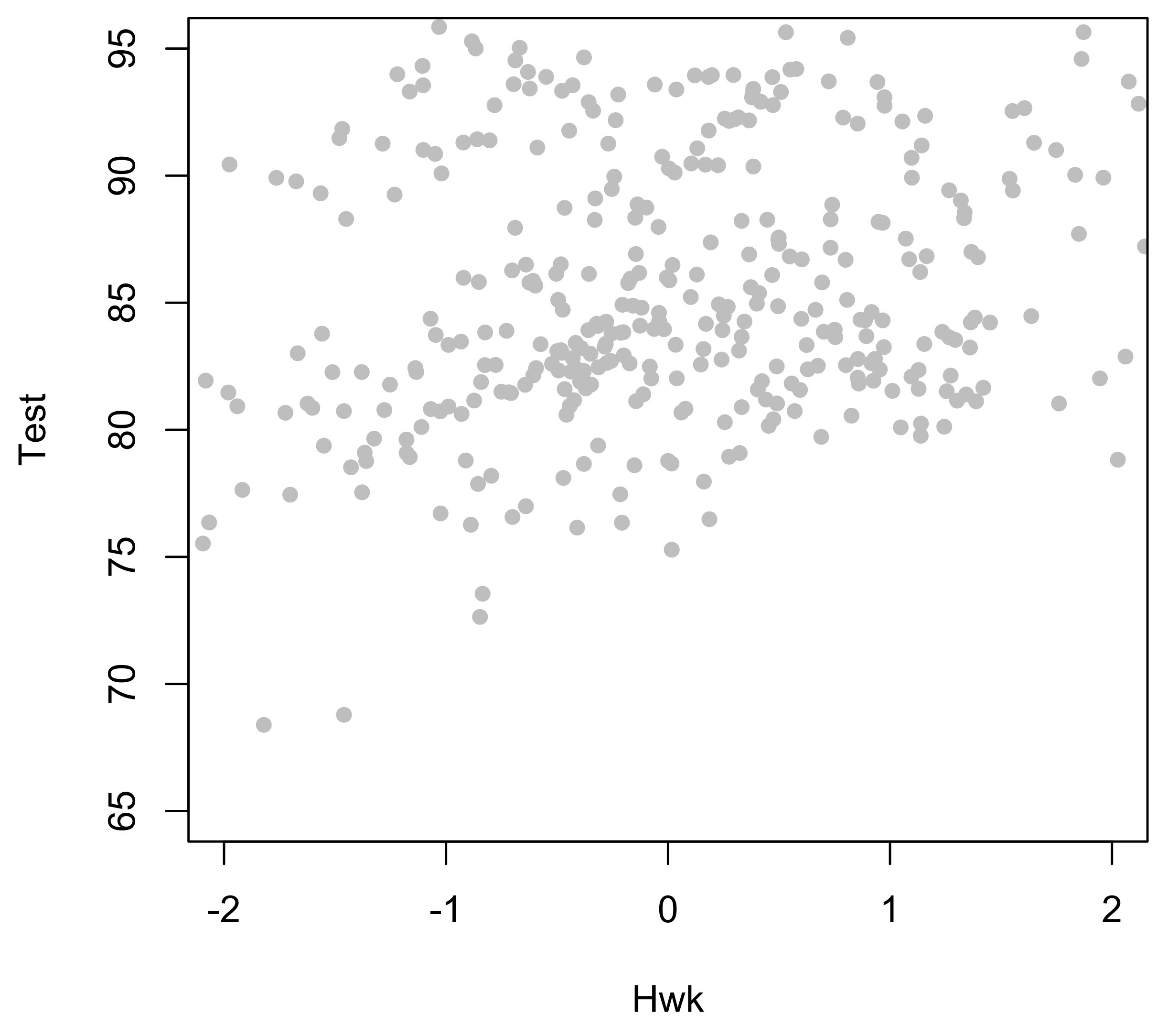
$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \qquad \epsilon_i \sim N(0, \sigma^2)$$



DEPENDENCE AS AN INTERESTING PHENOMENON

So now we found that we can get different EFFECTS of a covariate across schools!

So by aggregating across schools, we might actually be missing important information.



DEPENDENCE AS AN INTERESTIN ... ENON

So now we found that we can get d ... e across schools!

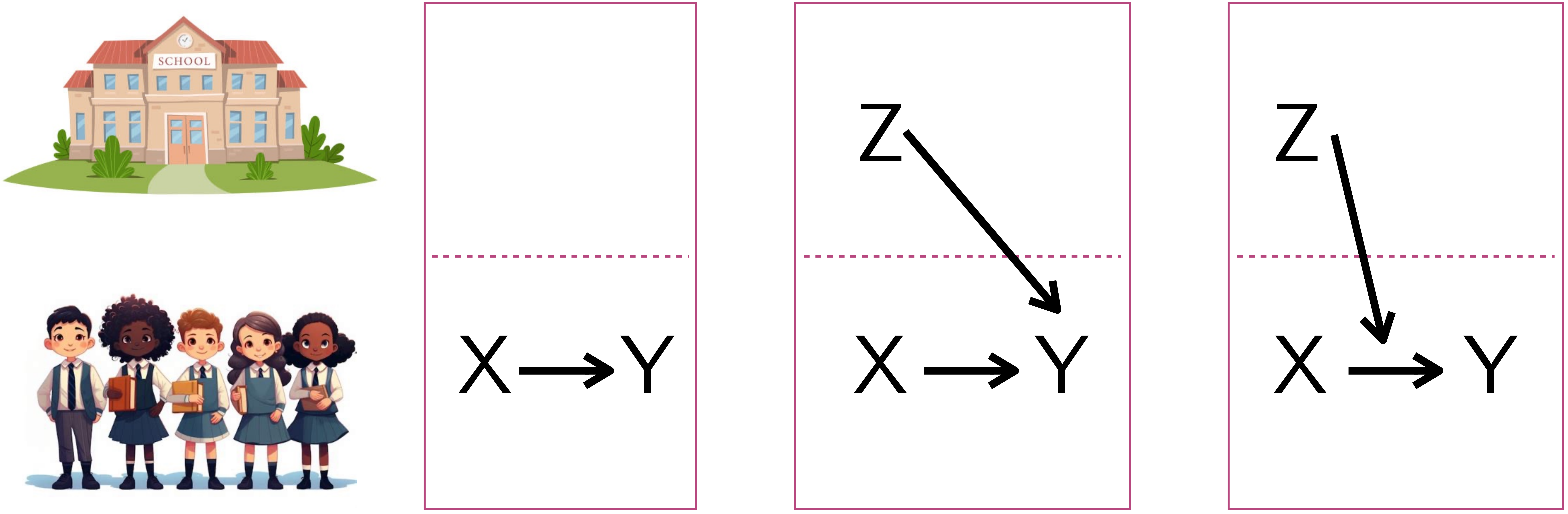
So by aggregating across schools, we m ... t information.

Reason #2 for Multilevel Models!

DEPENDENCE AS AN INTERESTING PHENOMENON

If we know that effects and intercepts vary across schools, we might be interested in answering research questions about why schools might vary.

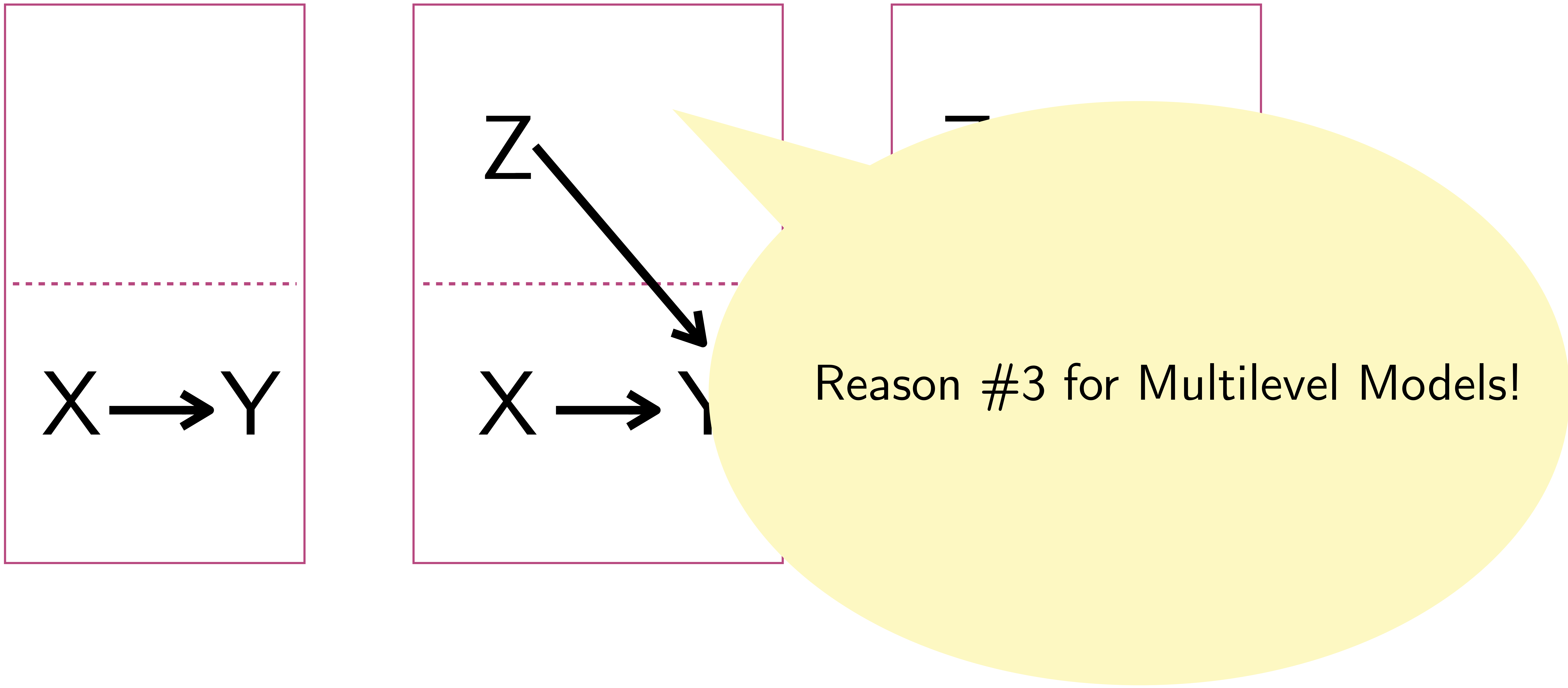
Multilevel Models allow us to answer research questions across multiple levels.



DEPENDENCE AS AN INTERESTING PHENOMENON

If we know that effects and intercepts vary across schools, we might be interested in answering research questions about why schools might vary.

Multilevel Models allow us to answer research questions across multiple levels.



OUR FIRST MULTILEVEL MODEL

$$Y_{ij} = \beta_{0j} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$u_{0j} \sim N(0,\ \tau_0^2)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

Also called the null model, the empty model, the unconditional model, the random intercept only model, etc.

Learn to recognize the equation!

FITTING MODELS IN R

There are two popular R packages for fitting multilevel models in R.

- LME4
- NLME

There is also other software for fitting multilevel models

- MPlus
- MLwiN

In addition to your standard statistical software

- SPSS
- SAS
- STATA

R CODE FOR LME4

We will use the LME4 R package.

```
install.packages(‘‘lme4’’)
library(lme4)
fit=lmer(outcome~1+(1|clusterID), data=   )
```

To see the random effects u_{0j}

```
ranef(fit)
```

To see the random intercepts β_{0j}

```
coef(fit)
```

R OUTPUT FOR LME4

$$Y_{ij} = \beta_{0j} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

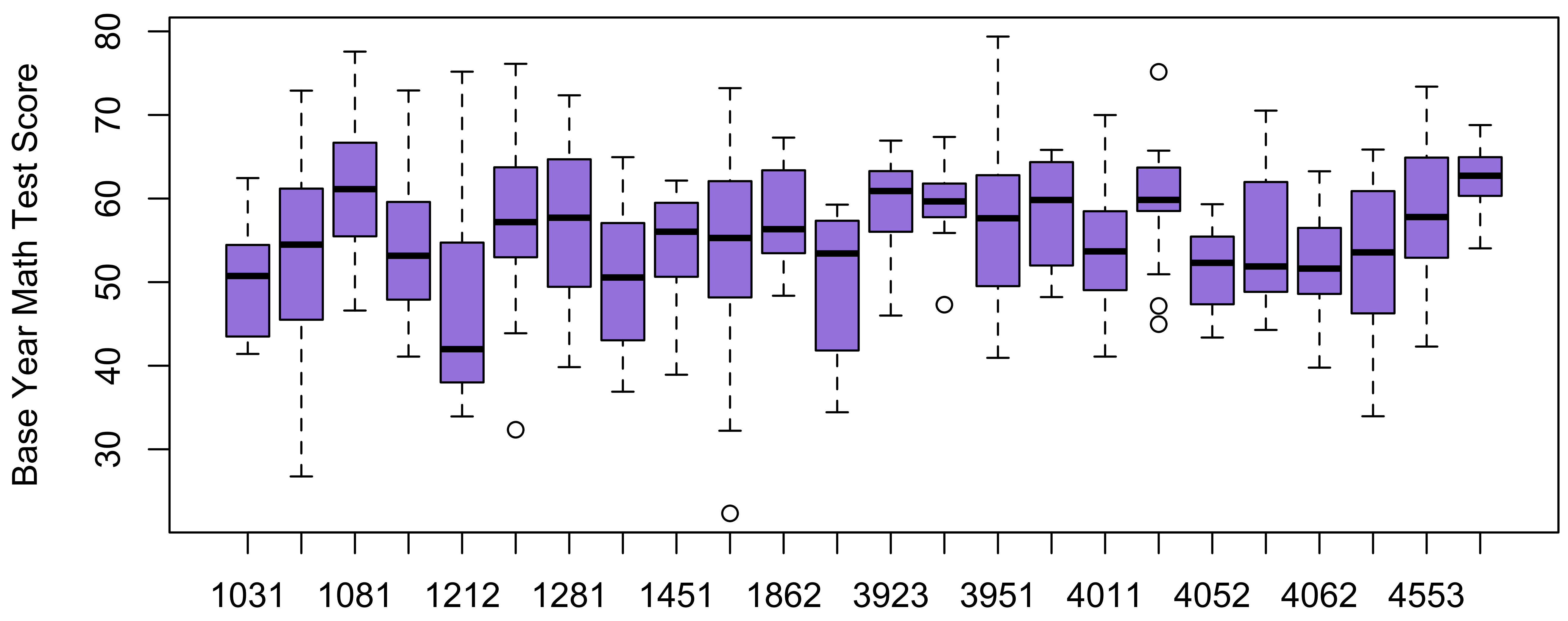
$$u_{0j} \sim N(0,\ \tau_0^2)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

```
Linear mixed model fit by REML ['lmerMod']
Formula: math ~ 1 + (1 | SCH_ID)
   Data: ELS

REML criterion at convergence: 22662.3

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.0698 -0.6213  0.0078  0.6618  3.3913

Random effects:
 Groups   Name        Variance Std.Dev.
 SCH_ID   (Intercept) 20.37    4.513
 Residual             69.29    8.324
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept)  53.4393     0.3146   169.9
```

WHY FIT A NULL MODEL?

TO GET THE ICC!



The Intraclass correlation (ICC) measures the proportion of variance that occurs between clusters relative to total amount of variance.

- ICC = Between cluster/ (Between cluster + within cluster)
- $\rho = \dfrac{\tau_0^2}{\tau_0^2 + \sigma^2}$

Large ICCs suggest a lot of dependence; near 0 ICCs suggest little dependence

CALCULATING THE ICC

$$Y_{ij} = \beta_{0j} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$u_{0j} \sim N(0,\ \tau_0^2)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

```
Random effects:
 Groups   Name        Variance Std.Dev.
 SCH_ID   (Intercept) 20.37    4.513
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Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept)  53.4393     0.3146   169.9
```

OUR SECOND MULTILEVEL MODEL

$$Y_{ij} = \beta_{0j} + \beta_1 X_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_1 = \gamma_{10}$$

$$u_{0j} \sim N(0,\ \tau_0^2)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

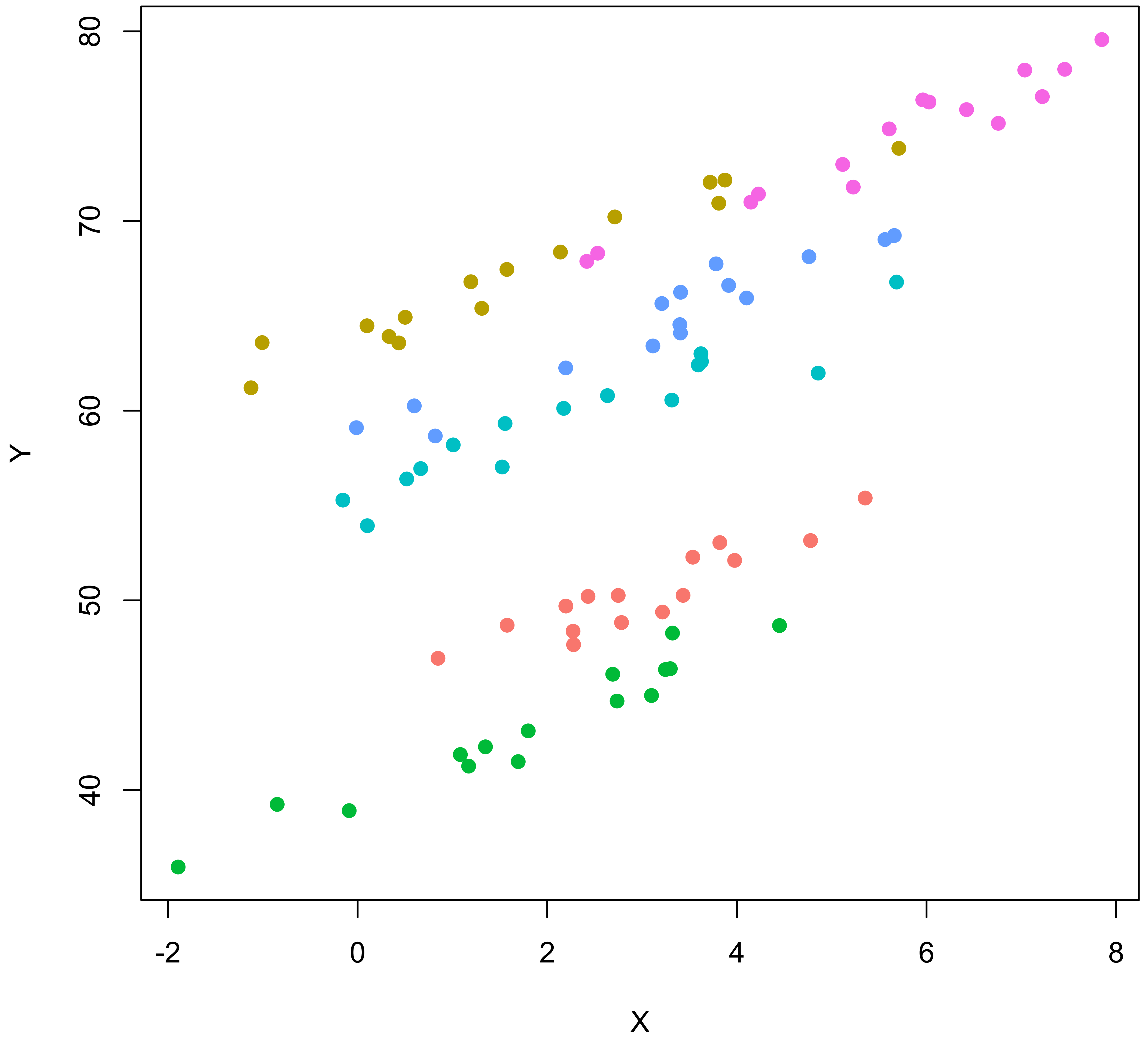
Also called the conditional model, the random effects ANCOVA model, the random intercept model, etc.

Learn to recognize the equation!

OUR SECOND MULTILEVEL MODEL

What does this model look like?

$$Y_{ij} = \beta_{0j} + \beta_1 X_{1ij} + \epsilon_{ij}$$
$$\beta_{0j} = \gamma_{00} + u_{0j}$$
$$\beta_1 = \gamma_{10}$$
$$u_{0j} \sim N(0,\ \tau_0^2)$$
$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$



FITTING THIS MODEL IN R

```
fit=lmer(outcome~1+X+(1|clusterID), data=  )

fit=lmer(outcome~X+(1|clusterID), data=  )
```

```
fit=lmer(math~1+hmwrk+(1|SCH_ID), data=ELS)

fit=lmer(math~hmwrk+(1|SCH_ID), data=ELS)
```

R OUTPUT

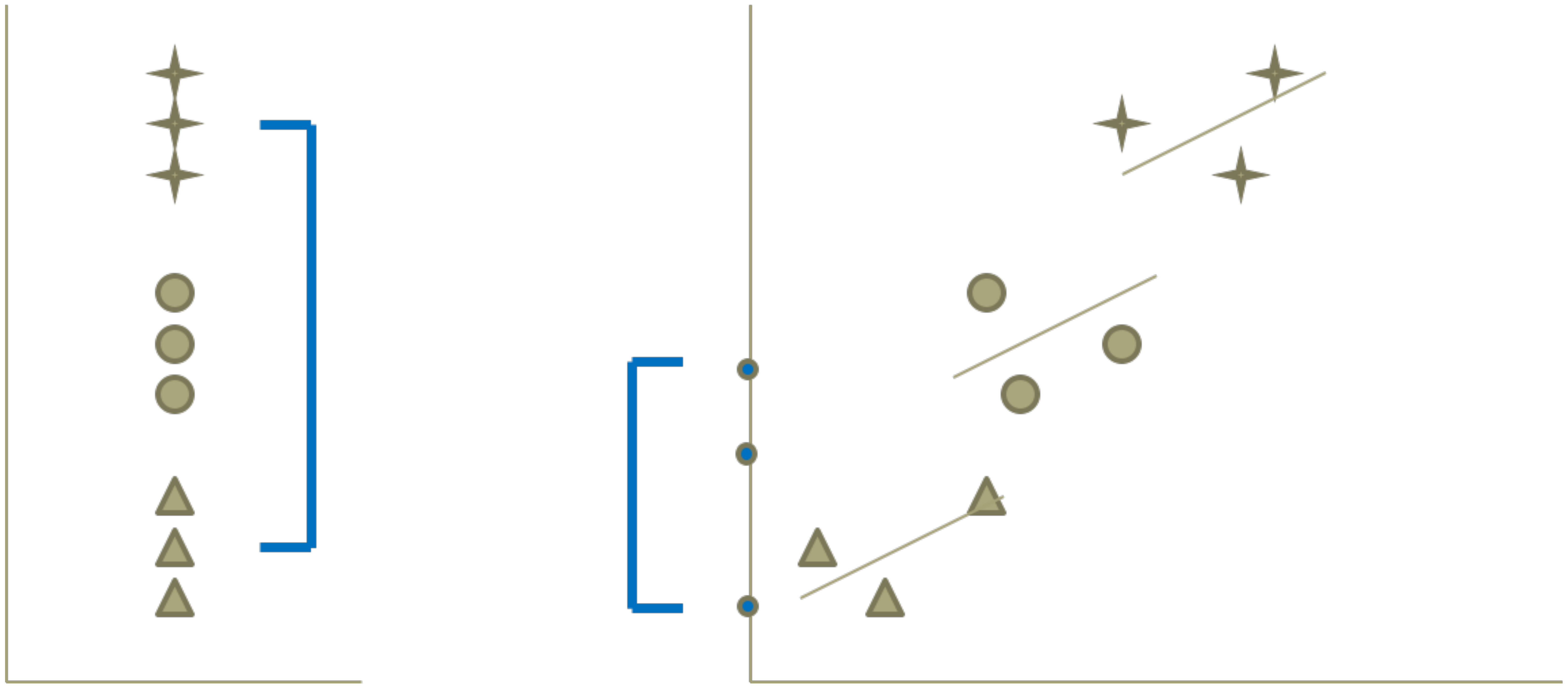
What happened to τ_0^2 when we added a level 1 variable?

```
Random effects:
 Groups   Name        Variance Std.Dev.
 SCH_ID   (Intercept) 17.06    4.131
 Residual             68.00    8.246
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept) 51.33850    0.36673 139.990
hmwrk        0.21007    0.02156   9.745
```

MORE ON ADDING COVARIATES

Adding a level 1 covariate may decrease our cluster level variance!



VARIANCE EXPLAINED

When predictors are added to the model (uncentered or grand mean centered), variability can be explained at the within and between levels (level 1 and level 2).

We can calculate change in R^2 at each level and combined:

Paremeter	Unconditional Model	Conditional Model
τ_0^2		
σ^2		

$$\frac{\sigma^2_{null} - \sigma^2_{new}}{\sigma^2_{null}}$$

Level 1

$$\frac{\tau^2_{0_{null}} - \tau^2_{0_{new}}}{\tau^2_{0_{null}}}$$

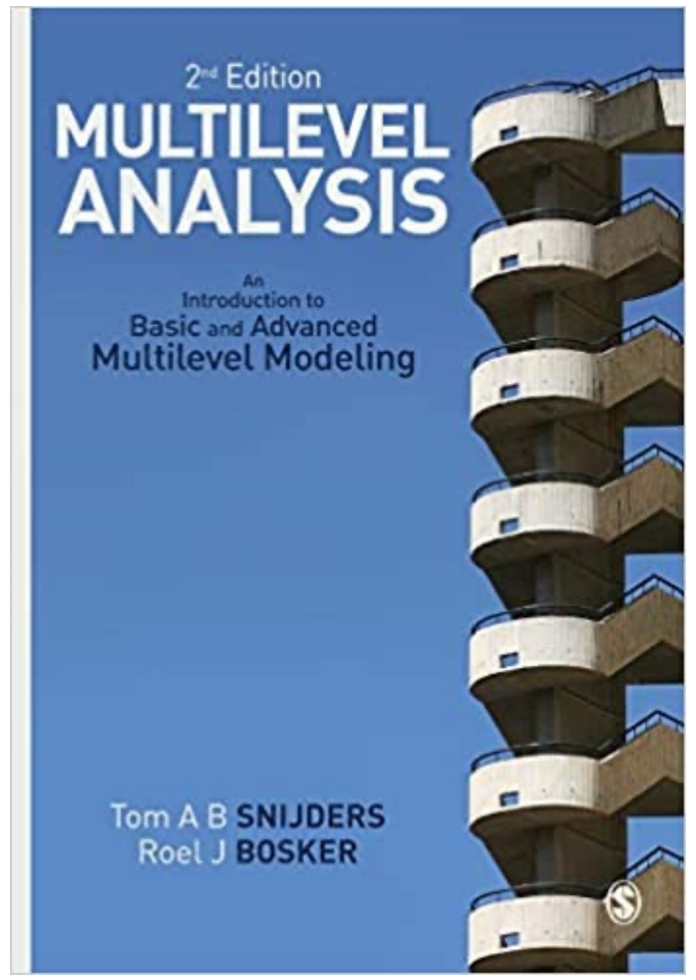
Level 2

$$\frac{\tau^2_{0_{null}} + \sigma^2_{null} - (\tau^2_{0_{new}} + \sigma^2_{new})}{\tau^2_{0_{null}} + \sigma^2_{null}}$$

Both

AN ASIDE: VARIANCE EXPLAINED IS STILL CHANGING

The definitions on the previous slide are from the Snijders and Bosker (2012) text.



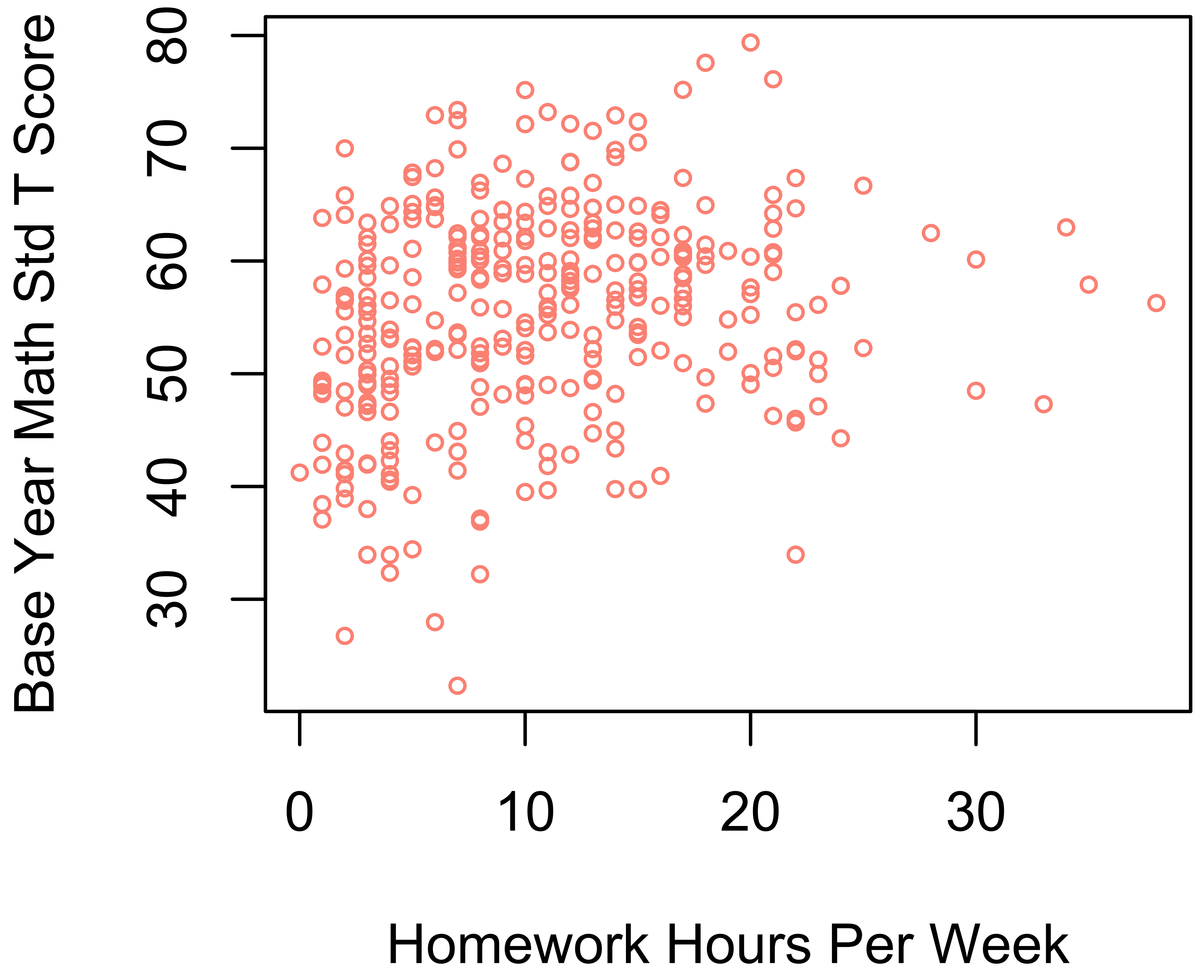
A recent paper about variance explained:

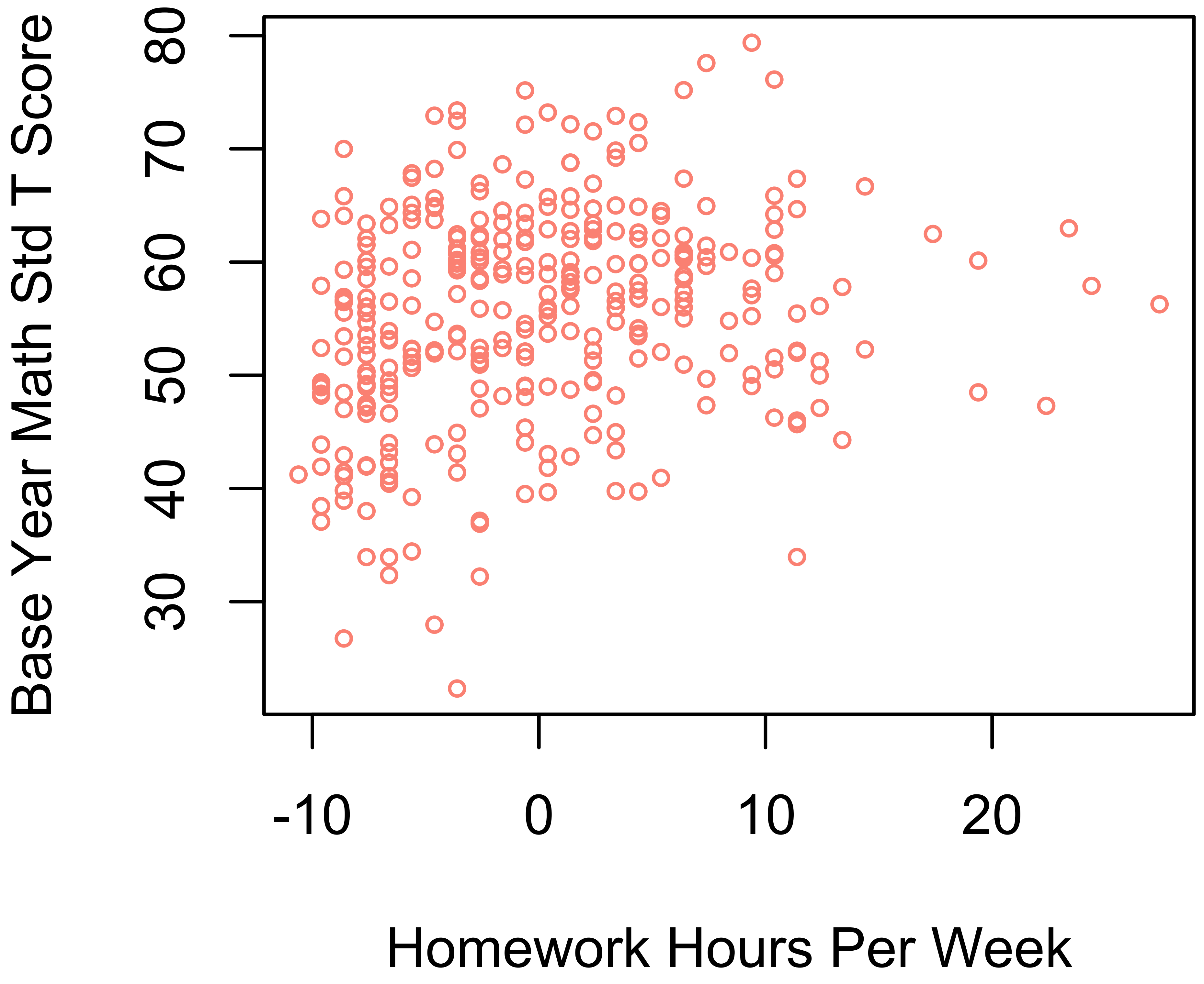
Rights, J. D., & Sterba, S. K. (2019). Quantifying explained variance in multilevel models: An integrative framework for defining R-squared measures. *Psychological Methods, 24*(3), 309–338. https://doi.org/10.1037/met0000184

CENTERING

GRAND MEAN CENTERING

Grand mean centering subtracts the covariate by the overall mean.
For multilevel data, this is the mean of all observations, regardless of cluster.





The plot doesn't change at all, just the scale of the x-axis!

GRAND MEAN CENTERING

Output without centering

```
Random effects:
 Groups   Name        Variance Std.Dev.
 SCH_ID   (Intercept) 17.06    4.131
 Residual             68.00    8.246
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept) 51.33850    0.36673 139.990
hmwrk        0.21007    0.02156   9.745
```

Output with grand-mean centering

```
Random effects:
 Groups   Name        Variance Std.Dev.
 SCH_ID   (Intercept) 17.06    4.131
 Residual             68.00    8.246
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept) 53.51806    0.29439 181.792
hmwrk.C      0.21007    0.02156   9.745
```

GROUP MEAN CENTERING

Grand Mean Centering

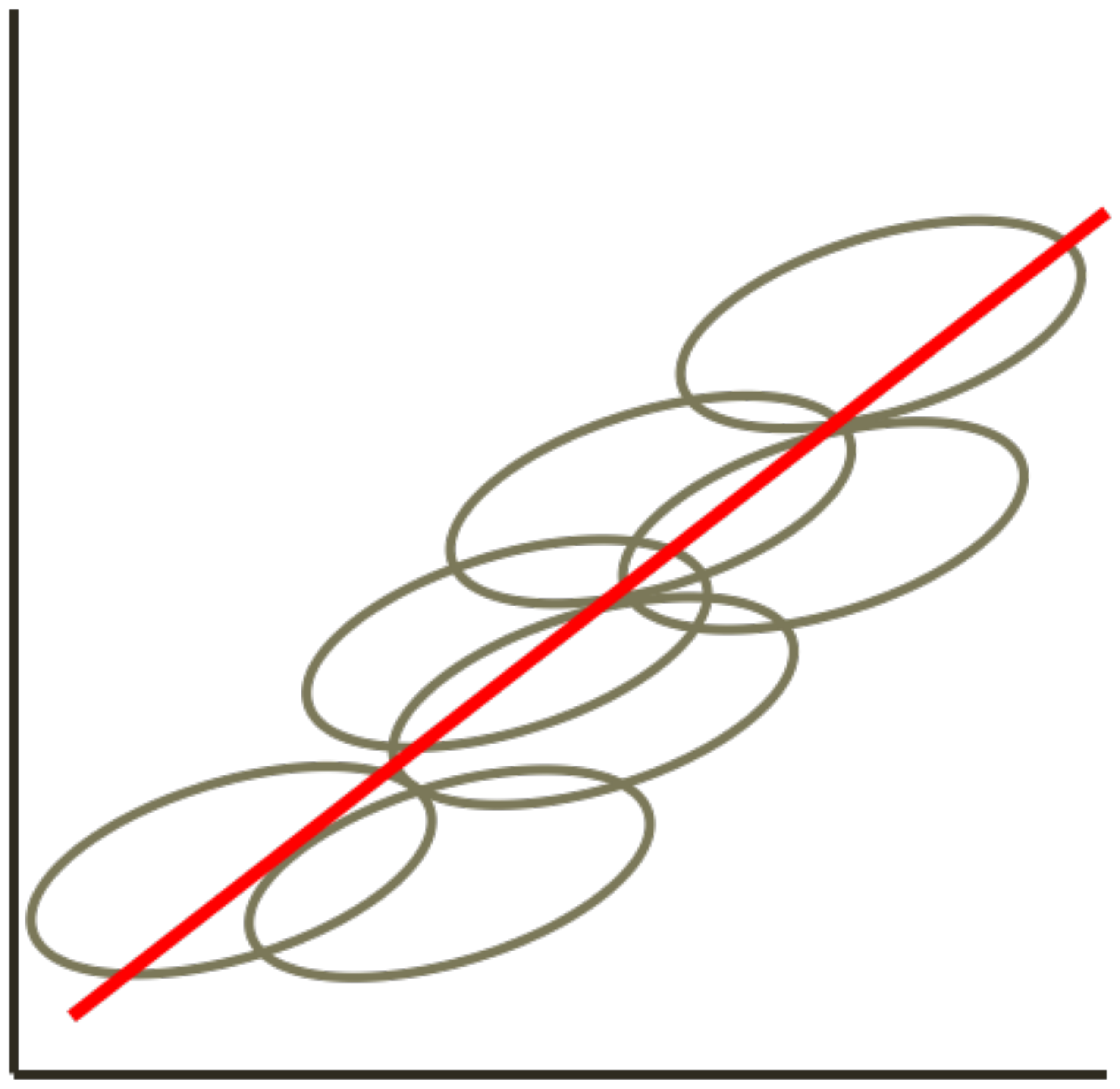
Subtract overall mean from each observation

$$X_{ij} - \overline{X}_{..}$$

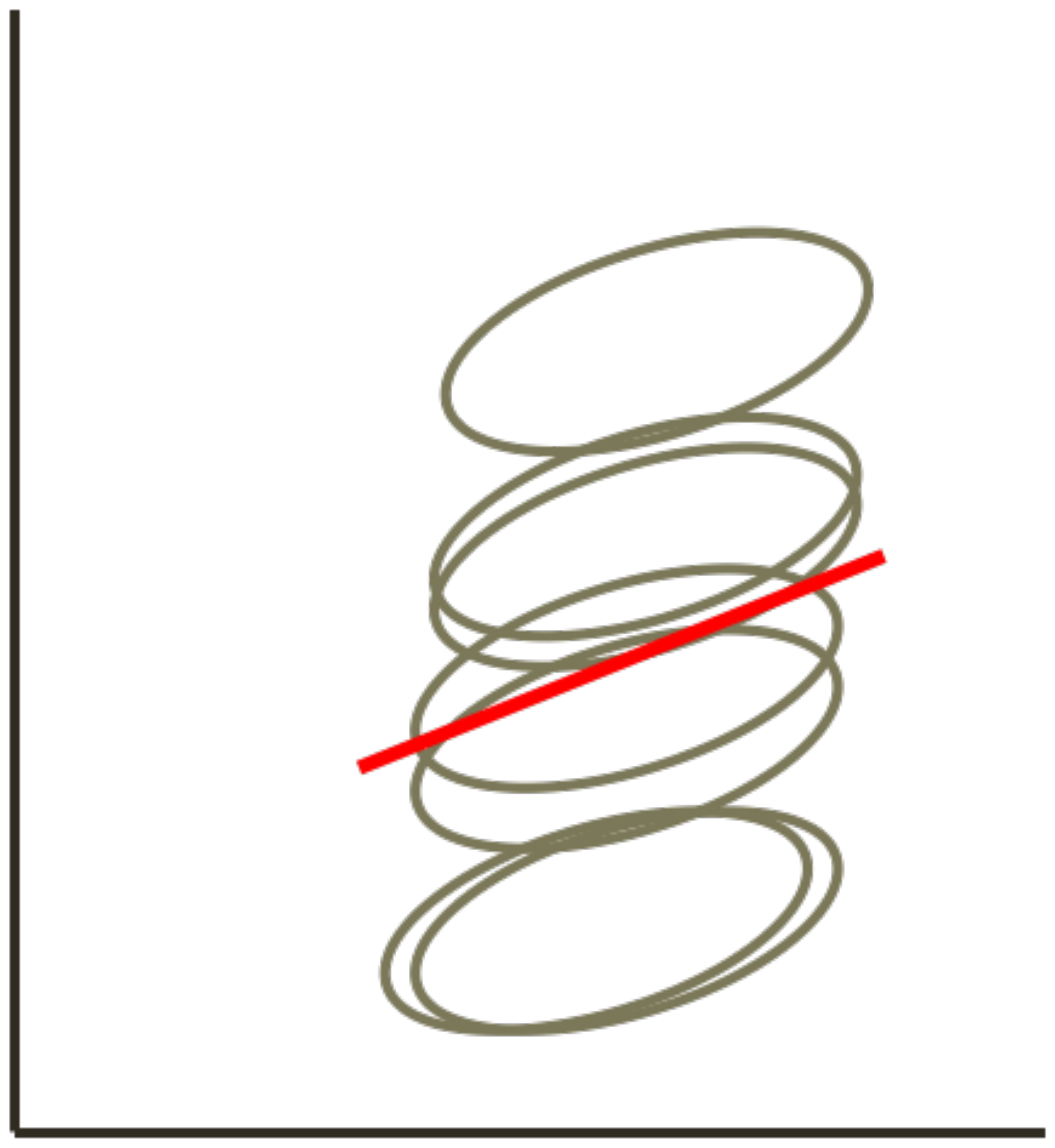
Group Mean Centering

Subtract the cluster mean from each observation in the cluster

$$X_{ij} - \overline{X}_{.j}$$



Relation without centering or with grand-mean centering



Relation with group-mean centering

HOW TO GROUP-MEAN CENTER IN R

You can use the group.mean function in the robumeta R package

```
install.packages(“robumeta”)
library(robumeta)
ELS$math.grpC=group.center(ELS$math, ELS$SCH_ID)
```

This has already been done for our ELS dataset, so use this code for future datasets.

INTERPRETING A GROUP-MEAN CENTERED PREDICTOR

Predicting math from hmwrk where hmwrk is group-mean centered

```
Random effects:
 Groups   Name        Variance Std.Dev.
 SCH_ID   (Intercept) 20.57    4.536
 Residual             67.91    8.241
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept) 53.43351    0.31491 169.679
hmwrk.grpC   0.17094    0.02227   7.675
```

$$Y_{ij} = \beta_{0j} + \beta_1(X_{1ij} - \overline{X_{.j}}) + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_1 = \gamma_{10}$$

$$u_{0j} \sim N(0,\ \tau_0^2)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

COMPARING GRAND-MEAN AND GROUP-MEAN CENTERING

Predicting math from hmwrk where hmwrk is **grand-mean centered**

```
Random effects:
 Groups   Name        Variance Std.Dev.
 SCH_ID   (Intercept) 17.06    4.131
 Residual             68.00    8.246
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept) 53.51806    0.29439 181.792
hmwrk.C      0.21007    0.02156   9.745
```

Predicting math from hmwrk where hmwrk is **group-mean centered**

```
Random effects:
 Groups   Name        Variance Std.Dev.
 SCH_ID   (Intercept) 20.57    4.536
 Residual             67.91    8.241
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept) 53.43351    0.31491 169.679
hmwrk.grpC   0.17094    0.02227   7.675
```

LEVEL 2 PREDICTORS

LEVEL 2 PREDICTORS FOR THE INTERCEPT

$$Y_{ij} = \beta_{0j} + \beta_1 X_{1ij} + \epsilon_{ij}$$

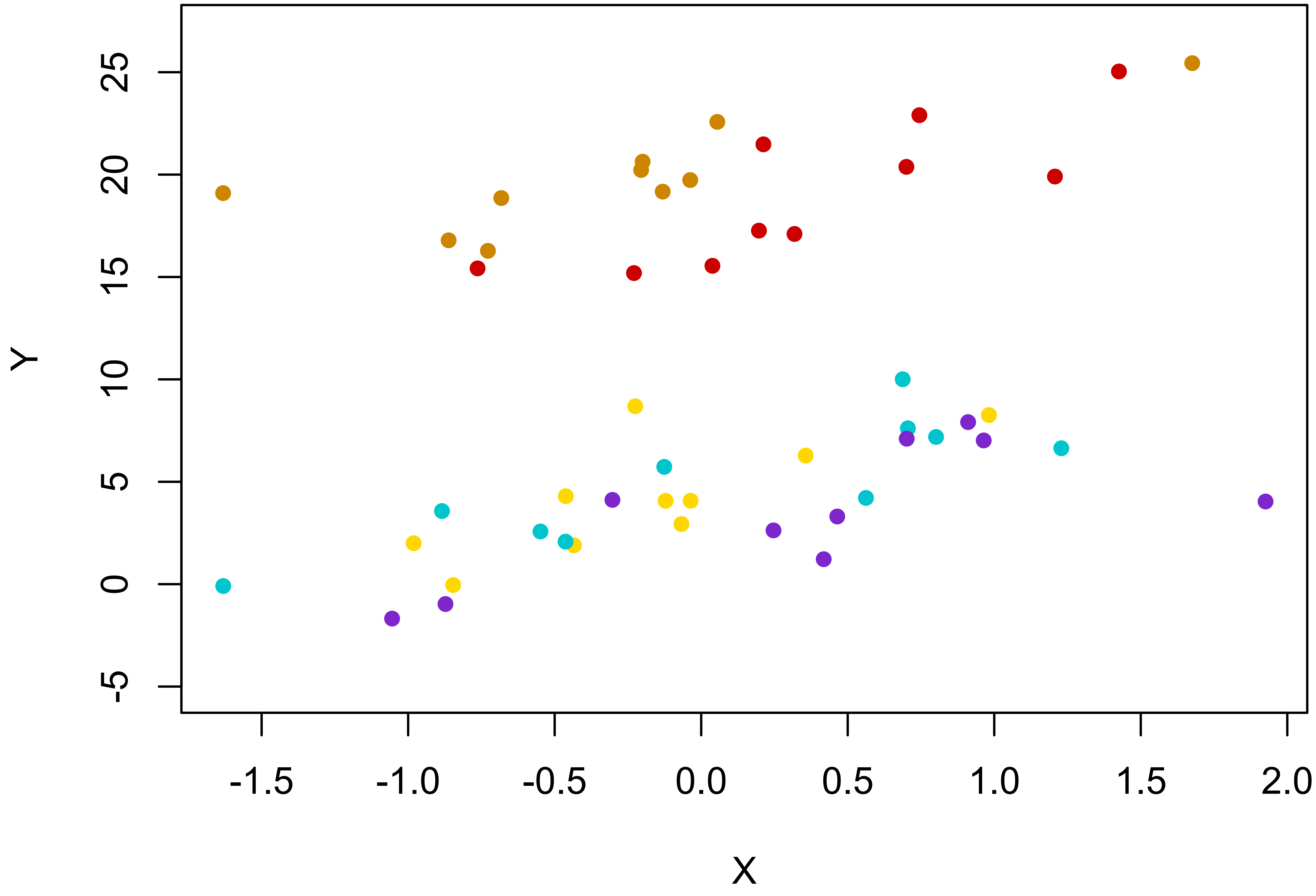
$$\beta_{0j} = \gamma_{00} + \gamma_{01} Z_j + u_{0j}$$

$$\beta_1 = \gamma_{10}$$

$$u_{0j} \sim N(0,\ \tau_0^2)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

A BINARY LEVEL 2 PREDICTOR



FITTING THIS MODEL IN R

```
fit = lmer(outcome~X+Z+(1|clusterID), data=MYDATA)
fit = lmer(math~hmwrk+private+(1 | SCH_ID), data=ELS)
```

$$Y_{ij} = \beta_{0j} + \beta_1 X_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01} Z_j + u_{0j}$$

$$\beta_1 = \gamma_{10}$$

$$u_{0j} \sim N(0,\ \tau_0^2)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

INTERPRETING OUR MODEL

$$Y_{ij} = \beta_{0j} + \beta_1 X_{1ij} + \epsilon_{ij}$$
$$\beta_{0j} = \gamma_{00} + \gamma_{01} Z_j + u_{0j}$$
$$\beta_1 = \gamma_{10}$$
$$u_{0j} \sim N(0,\ \tau_0^2)$$
$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

```
Random effects:
 Groups   Name        Variance Std.Dev.
 SCH_ID   (Intercept) 16.46    4.057
 Residual             68.04    8.248
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept) 51.86801    0.37923 136.773
hmwrk.grpC   0.17094    0.02229   7.668
private      3.90146    0.59042   6.608
```

CONTEXTUAL EFFECTS

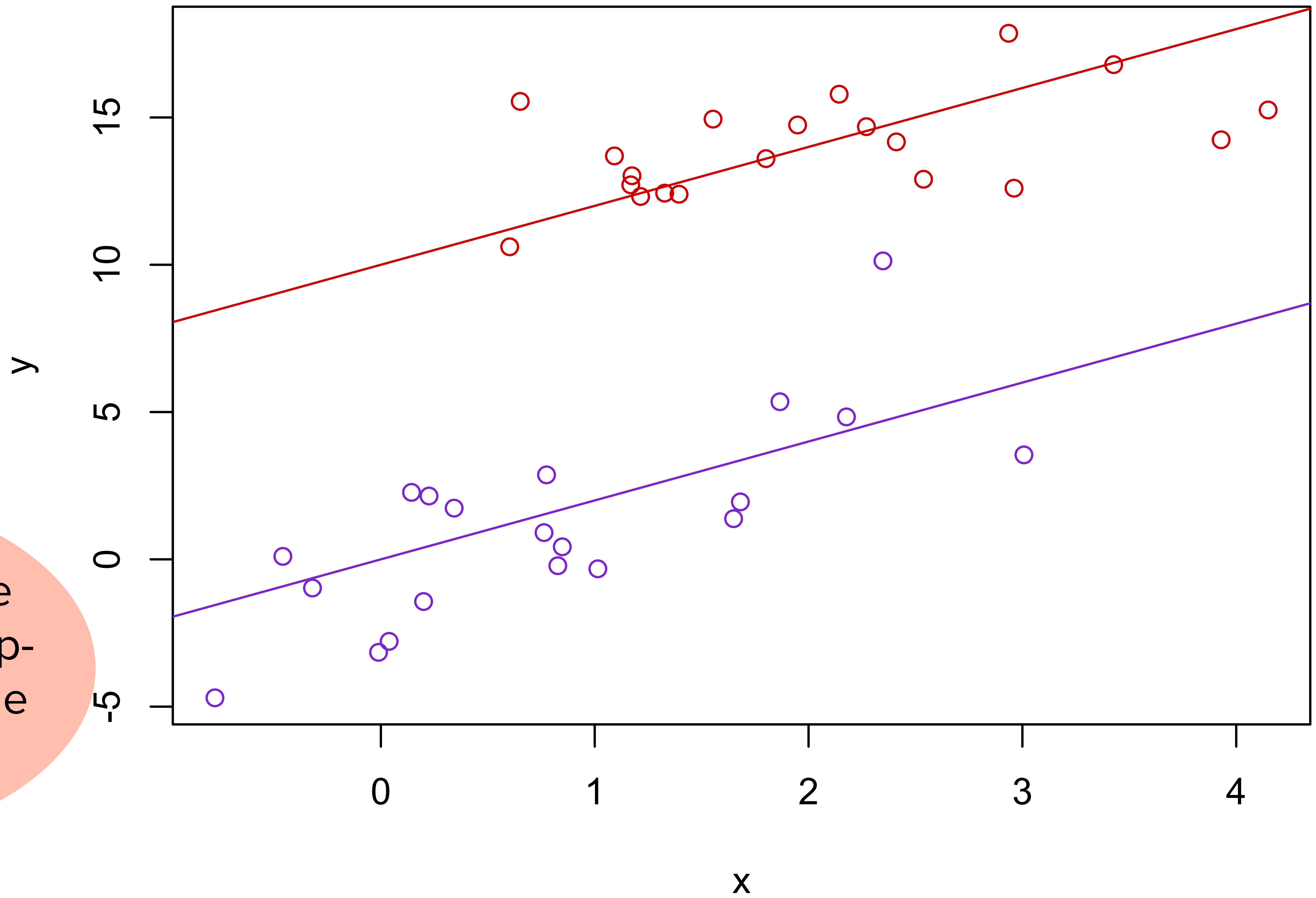
CONTEXT MATTERS!!

A student who does 1 hours of homework per night in a school that does an average of 2 hours per night versus a student who does 1 hour of homework per week in a school where the average is 1 hour.

A student who gets a B in math where the average person gets a C vs a student who gets a B in math where the average student gets a A.

MLM ALLOWS US TO ESTIMATE CONTEXTUAL EFFECTS

The contextual effect is the expected difference in outcomes between two observations, who have the same value for X but who are in clusters whose $\overline{X_{\cdot j}}$ (cluster means) differ by 1 unit.



Contextual effects are the effects of the group-level, controlling for the individual-level

WE CAN ALSO ESTIMATE THE CONTEXTUAL EFFECT

Grand Mean Centering

$$Y_{ij} = \beta_{0j} + \beta_1(X_{1ij} - \overline{X_{..}}) + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01}\overline{X_{.j}} + u_{0j}$$

$$\beta_1 = \gamma_{10}$$

$$u_{0j} \sim N(0, \tau_0^2)$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

γ_{01} is the contextual effect!

Group Mean Centering

$$Y_{ij} = \beta_{0j} + \beta_1(X_{1ij} - \overline{X_{.j}}) + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01}\overline{X_{.j}} + u_{0j}$$

$$\beta_1 = \gamma_{10}$$

$$u_{0j} \sim N(0, \tau_0^2)$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

$\gamma_{01} - \gamma_{10}$ is the contextual effect!

AN EXAMPLE TO DO TOGETHER

```
fit=lmer(math~hmwrk.grpC+Agg.hmwrk+(1|SCH_ID), data=ELS)
summary(fit)
```

When X is group mean centered the contextual effect is $\gamma_{01} - \gamma_{10}$!

```
Random effects:
 Groups   Name        Variance Std.Dev.
 SCH_ID   (Intercept) 13.51    3.676
 Residual             67.87    8.238
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept) 45.67004    0.84222  54.226
hmwrk.grpC   0.17094    0.02226   7.677
Agg.hmwrk    0.77208    0.07844   9.843
```

$$math_{ij} = \beta_{0j} + \beta_1 hmwrk\,.\,GrpC_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01} Agg\,.\,hmwrk + u_{0j}$$

$$\beta_1 = \gamma_{10}$$

$$u_{0j} \sim N(0,\ \tau_0^2)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

WHAT IF WE HAD GRAND MEAN CENTERED?

```
fit=lmer(math~hmwrk.C+Agg.hmwrk+(1|SCH_ID), data=ELS)
summary(fit)
```

When X is grand mean centered the contextual effect is γ_{01}!

```
Random effects:
 Groups   Name        Variance Std.Dev.
 SCH_ID   (Intercept) 13.51    3.676
 Residual             67.87    8.238
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept) 47.44359    0.87333  54.325
hmwrk.C      0.17094    0.02226   7.677
Agg.hmwrk    0.60114    0.08154   7.372
```

$$math_{ij} = \beta_{0j} + \beta_1 hmwrk\,.\,C_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01} Agg\,.\,hmwrk + u_{0j}$$

$$\beta_1 = \gamma_{10}$$

$$u_{0j} \sim N(0,\ \tau_0^2)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

MLMS WITH RANDOM SLOPES

RANDOM SLOPE MODELS

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

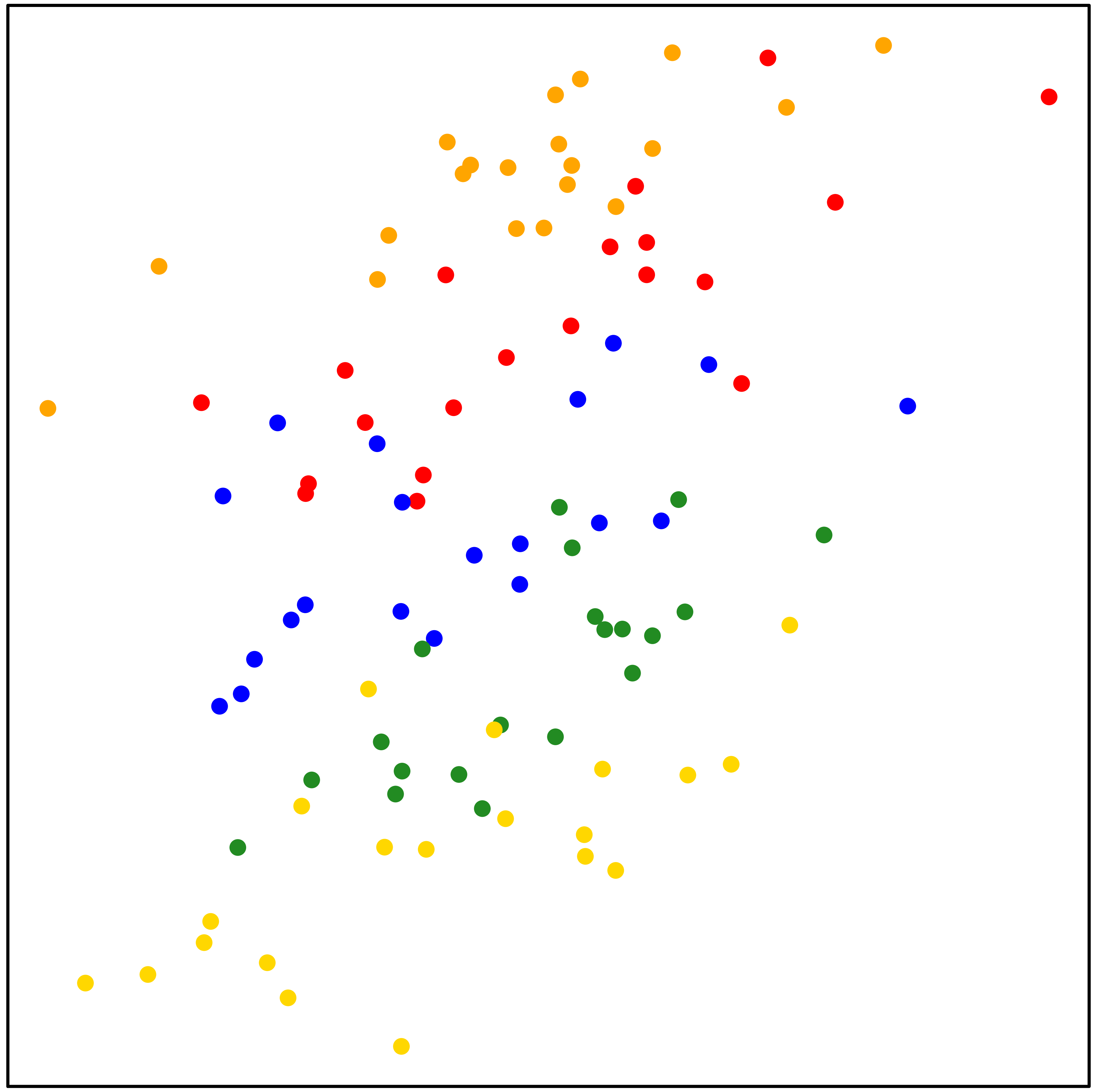
$$\beta_{1j} = \gamma_{10} + u_{1j}$$

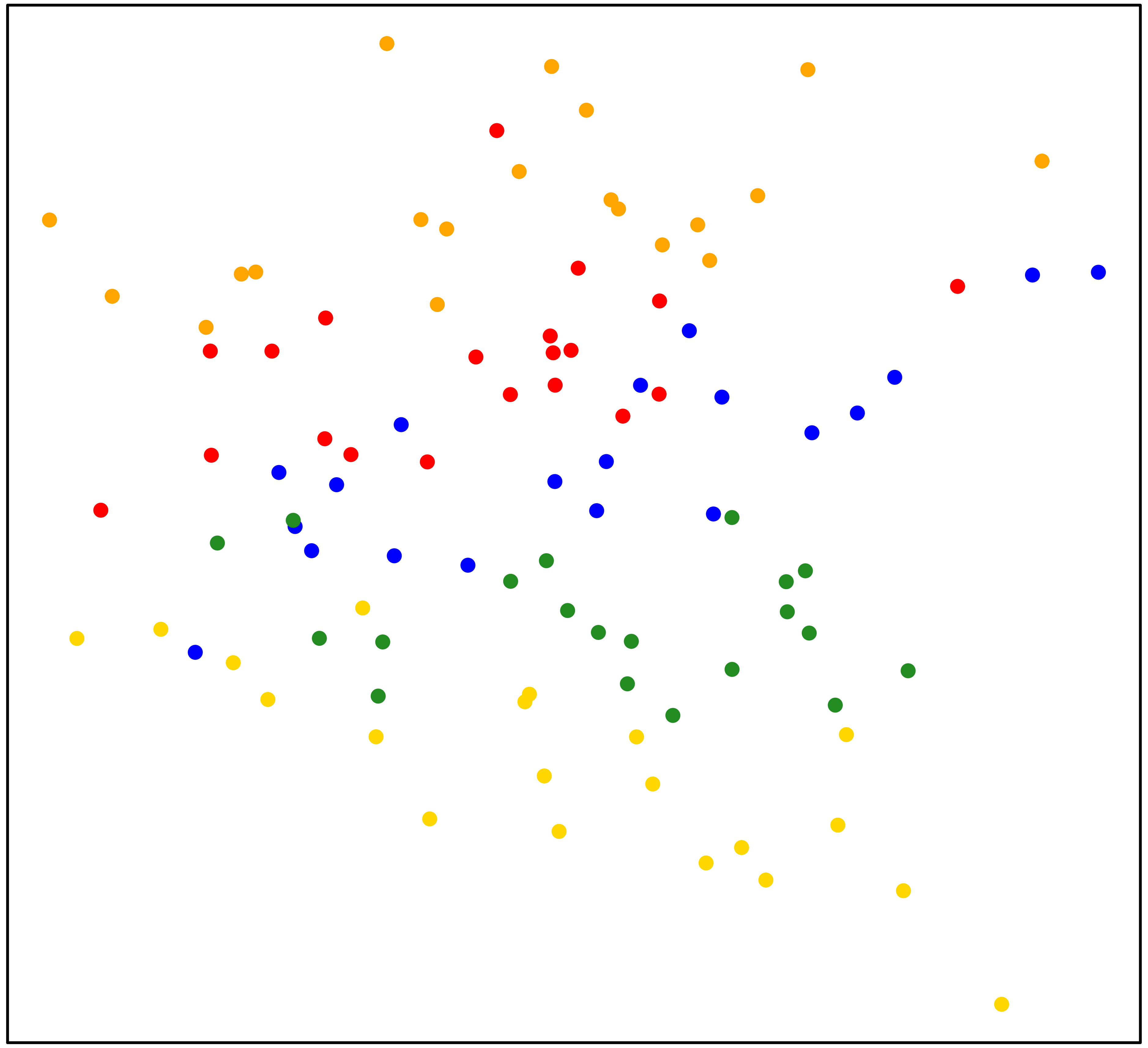
$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

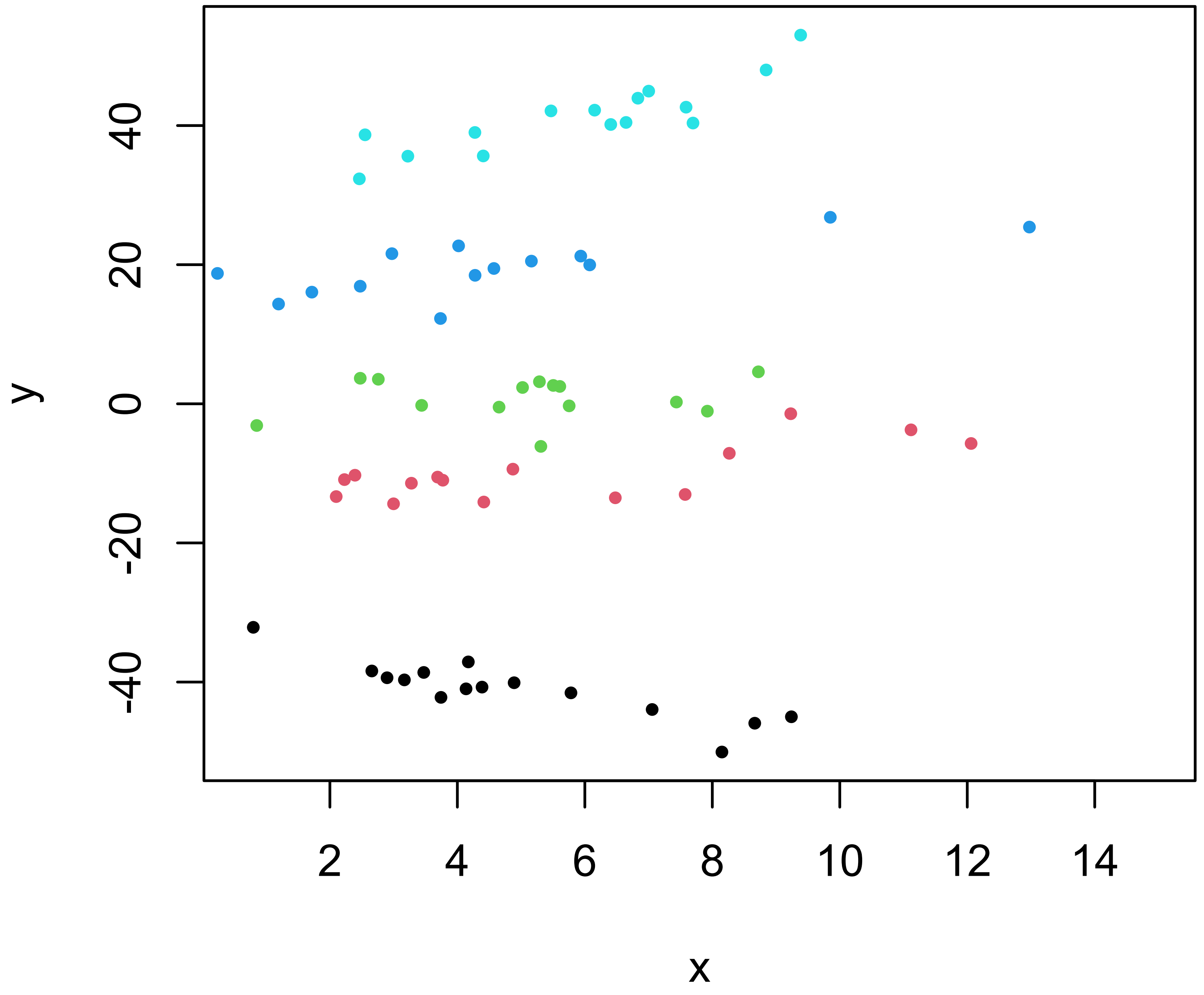
Note that random slopes contribute variance components about the variance in slope but also COVARIANCE components about the relation between random effects.

RANDOM INTERCEPT VS RANDOM INTERCEPT AND SLOPE

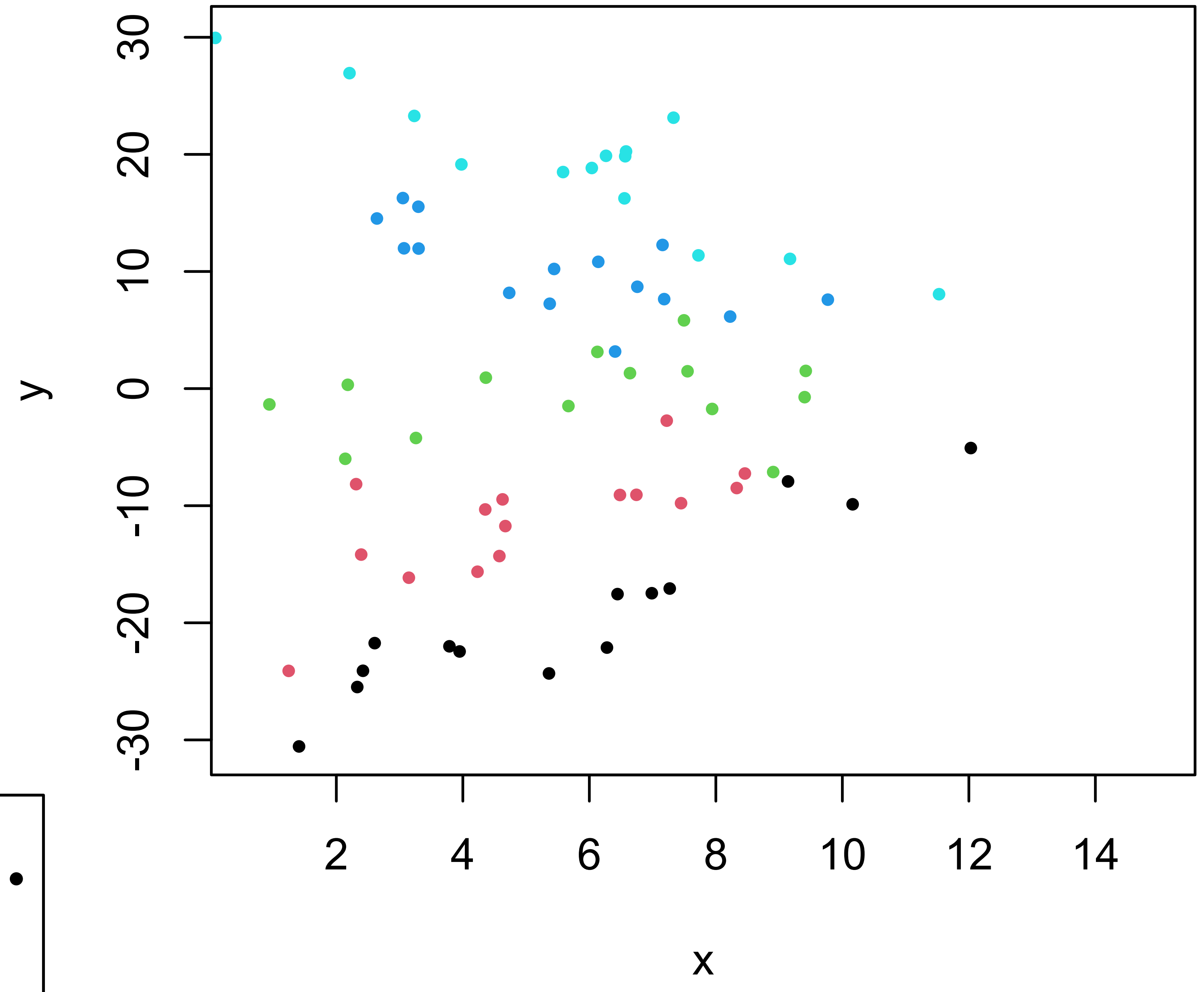


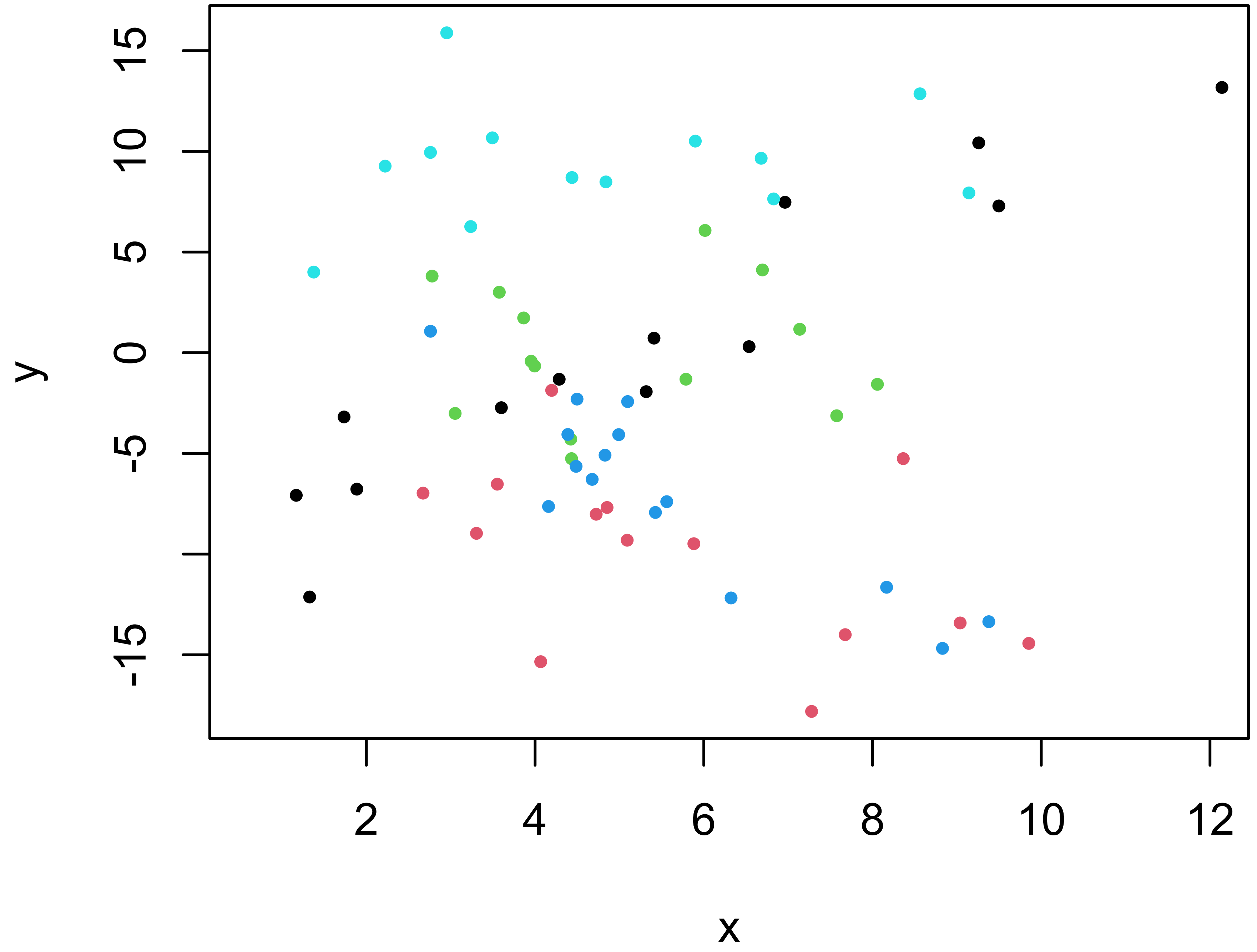


RANDOM INTERCEPT AND SLOPE COVARIANCE



$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right)$$





RANDOM SLOPE MODEL R CODE

```
fit=lmer(Y~1+X+(X|clusterID), data=mydata)
fit=lmer(math~hmwrk.grpC+(hmwrk.grpC|SCH_ID), data=ELS)
```

How to get the random effects u_{0j} and u_{1j}?

`ranef(fit)`

`coef(fit)` gives you β_{0j} and β_{1j}

How to get the tau matrix?

`as.data.frame(VarCorr(fit))`

RANDOM SLOPE OUTPUT

```
Random effects:
 Groups   Name        Variance Std.Dev. Corr
 SCH_ID   (Intercept) 20.71048 4.5509
          hmwrk.grpC   0.04313 0.2077   -0.37
 Residual             65.94742 8.1208
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
            Estimate Std. Error t value
(Intercept) 53.43518    0.31447  169.92
hmwrk.grpC   0.19216    0.02651    7.25
```

$$Y_{ij} = \beta_{0j} + \beta_{1j} X_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix} \right]$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

RANDOM SLOPE MODEL OUTPUT

How to get the tau matrix?

```
as.data.frame(VarCorr(fit))
```

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{1ij} + \epsilon_{ij}$$
$$\beta_{0j} = \gamma_{00} + u_{0j}$$
$$\beta_{1j} = \gamma_{10} + u_{1j}$$
$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right)$$
$$\epsilon_{ij} \sim N(0, \sigma^2)$$

	grp	var1	var2	vcov	sdcor
1	SCH_ID	(Intercept)	<NA>	20.71047702	4.5508765
2	SCH_ID	hmwrk.grpC	<NA>	0.04312925	0.2076758
3	SCH_ID	(Intercept)	hmwrk.grpC	-0.34980311	-0.3701201
4	Residual	<NA>	<NA>	65.94741936	8.1208016

PRACTICE WITH EQUATIONS

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right]$$

$$\epsilon_{ij} \sim N(0,\, \sigma^2)$$

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01}Z_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right]$$

$$\epsilon_{ij} \sim N(0,\, \sigma^2)$$

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{1ij} + \beta_2 X_{2ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\beta_2 = \gamma_{20}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right]$$

$$\epsilon_{ij} \sim N(0,\, \sigma^2)$$

PRACTICE WITH R CODE

```
fit=lmer(math~hmwrk.grpC+Agg.hmwrk+(hmwrk.grpC|SCH_ID), data=ELS)
```

```
fit=lmer(math~hmwrk+read+(hmwrk+read|SCH_ID), data=ELS)
```

```
fit=lmer(math~hmwrk+riskfc+read+(read|SCH_ID), data=ELS)
```

LEVEL 2 PREDICTORS PREDICTING SLOPES

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{1ij} + \epsilon_{ij}$$
$$\beta_{0j} = \gamma_{00} + \gamma_{01}Z_j + u_{0j}$$
$$\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right)$$
$$\epsilon_{ij} \sim N(0,\, \sigma^2)$$

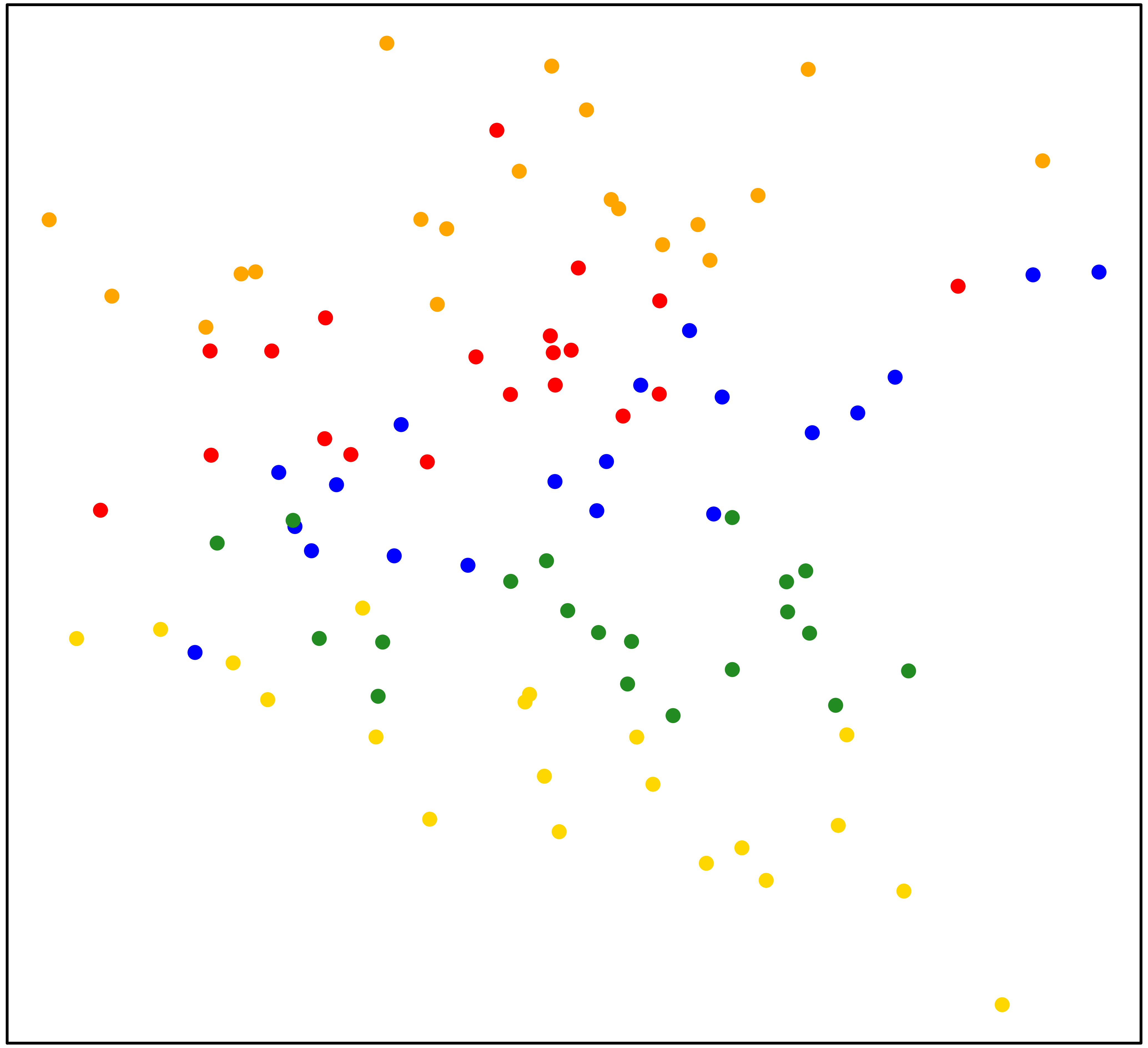
$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{1ij} + \epsilon_{ij}$$
$$\beta_{0j} = \gamma_{00} + u_{0j}$$
$$\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right)$$
$$\epsilon_{ij} \sim N(0,\, \sigma^2)$$

Note: These are sometimes called slopes as outcomes models.

LEVEL 2 VARIABLE PREDICTING SLOPE

Reminder: We must have different slopes across clusters if we want a level 2 variable explaining it!



LEVEL 2 PREDICTORS PREDICTING SLOPES

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01}Z_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix} \right)$$

$$\epsilon_{ij} \sim N(0,\, \sigma^2)$$

The tricky thing about these models is that they are actually models with interaction effects.

LEVEL 2 VARIABLE PREDICTING RANDOM SLOPE MODEL IN R

```
fit = lmer(Y ~ X+Z+X*Z+(X|clusterID), data=mydata)

fit = lmer(math ~ hmwrk+private+hmwrk*private+(hmwrk|SCH_ID),
data=ELS)
```

Note that this code will automatically include Z to predict the random intercept AND the random slope.

```
fit = lmer(Y ~ X*Z+(X|clusterID), data=mydata)
```

If you want to only include Z as the predictor for the random slope and not the intercept, use this code:

```
fit = lmer(Y ~ 1 + X *Z - Z + (X |clusterID), data=mydata)
```

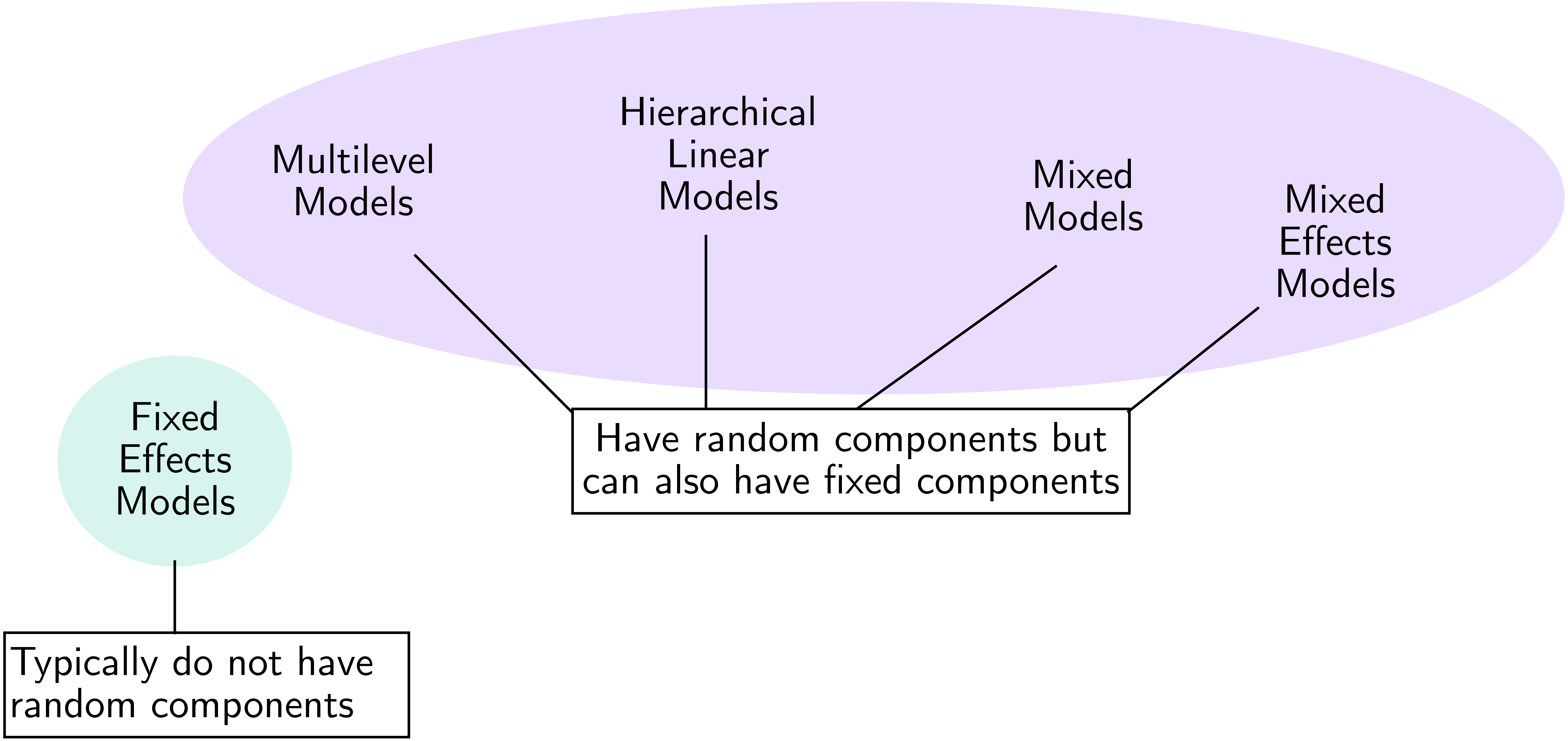
INTERPRETING R OUTPUT

```
Random effects:
 Groups   Name        Variance Std.Dev. Corr
 SCH_ID   (Intercept) 16.72147 4.0892
          hmwrk.grpC   0.03416 0.1848   -0.18
 Residual             66.08618 8.1293
Number of obs: 3148, groups:  SCH_ID, 291

Fixed effects:
                   Estimate Std. Error t value
(Intercept)        51.87145    0.37964 136.635
hmwrk.grpC          0.26156    0.03422   7.644
private             3.88568    0.59124   6.572
hmwrk.grpC:private -0.17916    0.05202  -3.444
```

FIXED EFFECTS VS RANDOM EFFECTS

FIXED EFFECTS VS RANDOM EFFECTS MODELS



FIXED EFFECTS VS RANDOM EFFECTS MODELS

Multilevel Models

Hierarchical Linear Models

Mixed Models

Mixed Effects Models

Have random components but can also have fixed components

Fixed Effects Models

Typically do not have random components

Both of these models are for multilevel data!

FIXED EFFECTS VS RANDOM EFFECTS

Fixed effects are indicator variables to represent each cluster's intercept (or slope).

$$Y_{ij} = \beta_0 + C_1 + C_2 + C_3 + \ldots + C_p + \beta_1 X_{1ij} + \epsilon_{ij}$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

Random Effects are random intercepts and random slopes (we'll get those later).

$$Y_{ij} = \beta_{0j} + \beta_1 X_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_1 = \gamma_{10}$$

$$u_{0j} \sim N(0,\ \tau_0^2)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

HOW DO I CHOOSE?

Do your clusters represent the entire population?

Or are your clusters samples from a population?

- If you have fewer than 10 clusters, use fixed effects.
- If you have more than 30 clusters, use random effects.

MODEL SELECTION

BUILDING A SEQUENCE OF NESTED MODELS: LEVEL 1

Recommended procedures from Raudenbush & Bryk, 2001; Snijders & Bosker, 2012

1. Run an unconditional/null model
2. Add in level-1 predictors, carefully considering centering decisions
3. Consider interactions among level-1 predictors
4. Evaluate whether slopes vary across clusters. You may need to do this for one slope at a time.
5. (Some people will drop level 1 predictors at this point)
6. Check the assumptions of your level 1 model.

BUILDING A SEQUENCE OF NESTED MODELS: LEVEL 2

Recommended procedures from Raudenbush & Bryk, 2001; Snijders & Bosker, 2012

7. Add level 2 predictors to the intercept
8. Add level 2 predictors to those slopes modeled as random.
9. Consider adding the same level 2 predictors.
10. Determine the proportion of explained variance for each random effect (do this with the prediction of one random slope at a time)
11. Determine whether to retain your random slopes (based on model comparison or slope variances near 0).
12. Check the assumptions of your level 2 model.

HOW DO WE COMPARE MODELS?

Likelihood Ratio Test

- The ratio of likelihoods follows a chisq distribution.
- We can determine a better fitting model with a significant p-value <0.05.
- LRT can only be used on nested models, where a model can be written within another model.

AIC or BIC

- Akaike or Bayesian Information Criteria can compare nested or non-nested models.
- A difference of 2 is a rule of thumb to determine a difference in models.
- Both AIC and BIC penalize models for having many parameters, but BIC has a stronger penalty

LIKELIHOOD RATIO TEST

- LRT allows us to compare nested models through a significant test.
- A ratio of two nested loglikelihoods follows as χ^2 distribution.
- $-2\log\frac{L_A}{L_B} = -2\log L_A - (-2)\log L_B$ is the difference in Deviance.
- The null hypothesis is that this ratio is 1, so $-2\log(1) = 0$
- A p-value<0.05 suggests there is a difference in the likelihoods, so the more complicated model is better!
- A nonsignificant p-value suggests the models are equivalent, so we would choose the simpler model.

INFORMATION CRITERIA

- Information criteria (AIC, BIC, etc) allow us to compare non-nested models.
- A difference in AIC or BIC of about 2 is considered indicative of a better fitting model.
- Lower AIC and BIC is generally better, but check your statistical software.
- Akaike Information Criterion is sometimes defined as $-2\log L + 2p$ where p is the number of parameters in the model
- Bayesian Information Criterion is sometimes defined as $-2\log L + \log(n)p$ where p is the number of parameters in the model and n is the number of observations
    - BIC has a more severe penalty for models with many parameters

USE CAUTION WHEN COMPARING MODELS

- Statistical software varies in how they calculate Deviance, AIC, and BIC. Use functions consistently.
  - e.g., Use `AIC()` and `BIC()` instead of `extractAIC()` and `BIC()`
- The two models MUST be fit on the same dataset! This becomes a huge issue with missing data.
  - If Model 1 has variables X1 with missing values, and Model 2 has variables X2 with missing values, then these models are no longer comparable. You must address missing data prior to do doing model selection.

ASSESSING MODEL ASSUMPTIONS

MODEL ASSUMPTIONS

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{10}Z_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix} \right)$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

1. $\epsilon_{ij} \sim N(0, \sigma^2)$ implies that the level 1 residual errors are normally distributed, independent, with constant variance.
   - Level 1 residuals are independent
   - Level 1 residuals are normally distributed
   - Level 1 residuals are homoscedastic

2. $Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + \epsilon_{ij}$ specifies a linear relation (linearity assumption)

MODEL ASSUMPTIONS

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{10}Z_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix} \right)$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

3. $\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix} \right)$ so level 2 residuals are normally distributed, independent with constant variance

- Level 2 residuals/Random effects (intercepts and slopes) are normally distributed
- Level 2 residuals/Random effects from different clusters are independent
- Level 2 residuals/Random effects have constant variance

4. $\beta_{0j} = \gamma_{00} + \gamma_{01}Z_j + u_{0j}$ and $\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j}$ are both linear models (so linearity assumption).

MODEL ASSUMPTIONS

5. Level 1 and level 2 residuals are independent.

6. The predictors at each level are not correlated with the residuals at the other level. That is, level 1 predictors are independent from level 2 residuals are vice versa.

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + \epsilon_{ij}$$
$$\beta_{0j} = \gamma_{00} + \gamma_{10}Z_j + u_{0j}$$
$$\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right)$$
$$\epsilon_{ij} \sim N(0, \sigma^2)$$

HOW TO CHECK MODEL ASSUMPTIONS

- Assess independence, linearity, normality, and homoscedasticity of level 1 residuals
- Assess independence, linearity, normality, and homoscedasticity of level 2 residuals
- Assess independence (lack of correlation) between level 1 and level 2 residuals
- Assess independence (lack of correlation) between level 1 residuals and level 2 predictors AND level 2 residuals and level 1 predictors.

You may also want to check for influential points!

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{10}Z_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix} \right)$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

WHAT HAPPENS IF WE VIOLATE ASSUMPTIONS?

- Violating the **distributional** assumptions of level 1 and level 2 residuals (i.e. normality, independence, homoscedasticity) and independence of level 1 and level 2 residuals results in <u>bias in the standard errors of our parameter estimates (γ)</u>.
  - Why? These are the assumptions about the random effects in the model.
- Violating the **linearity assumptions** or the assumptions about **independence** between level 1 residuals and level 2 predictors and between level 2 residuals and level 1 predictors results in <u>bias in the point estimates of our parameter estimates (γ)</u>.
  - Why? These assumptions deal with model specification.

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{10}Z_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11}Z_j + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix} \right)$$

$$\epsilon_{ij} \sim N(0,\sigma^2)$$

R PACKAGE FOR DIAGNOSTICS

We will be using the HLMdiag package.

See https://cran.r-project.org/web/packages/HLMdiag/vignettes/hlm_resid.html for more information and some practice

We will do a brief demo of this R package!

TIPS FOR ASSESSING LEVEL 1 RESIDUALS

These tips come from a paper by Hilden-Mintro (1995) that is summarized in Snijders & Bosker (2012).

- Level 1 residuals come from within-cluster regression at level 1.
- Plot unstandardized within-cluster OLS residuals against level 1 predictors to check for linearity
- Use the standardized OLS residuals to check the normality assumption with a qqplot and Shapiro-Wilk test.
-

TIPS FOR ASSESSING LEVEL 2 RESIDUALS

- Level 2 residuals will always be confounded with level 1 residuals, so fix things at level 1 before going to level 2.
  - Level 2 residuals can be treated in the same way as you learned in previous courses
    - Plot unstandardized level 2 residuals against level 2 predictors (or fitted values) for linearity and homoescedasticity
    - Plot standardized level 2 residuals in a qq plot with Shapiro-Wilk test to examine normality
- BUT often, violations in assumptions are due to something wrong at level 1.

LONGITUDINAL MODELS

MULTILEVEL MODELS FOR LONGITUDINAL DATA

- Longitudinal data comes from multistage sampling too!
- We sample people and then sample observations within people.
- We have observations/ measurement times nested in people.

- Multilevel models are not the same as repeated measures ANOVA.
- The datasets are often the same, but MLM is much more flexible than repeated measures ANOVA.
- MLM does not require fixed measurement points (although we have them)
- MLM allows for missing measurements
- MLM allows for non-linear growth patterns

THE NULL MODEL

$$Y_{ij} = \beta_{0j} + \epsilon_{ij}$$
$$\beta_{0j} = \gamma_{00} + u_{0j}$$
$$u_{0j} \sim N(0,\ \tau_0^2)$$
$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

where Y_{ij} is the outcome for measurement i for person j

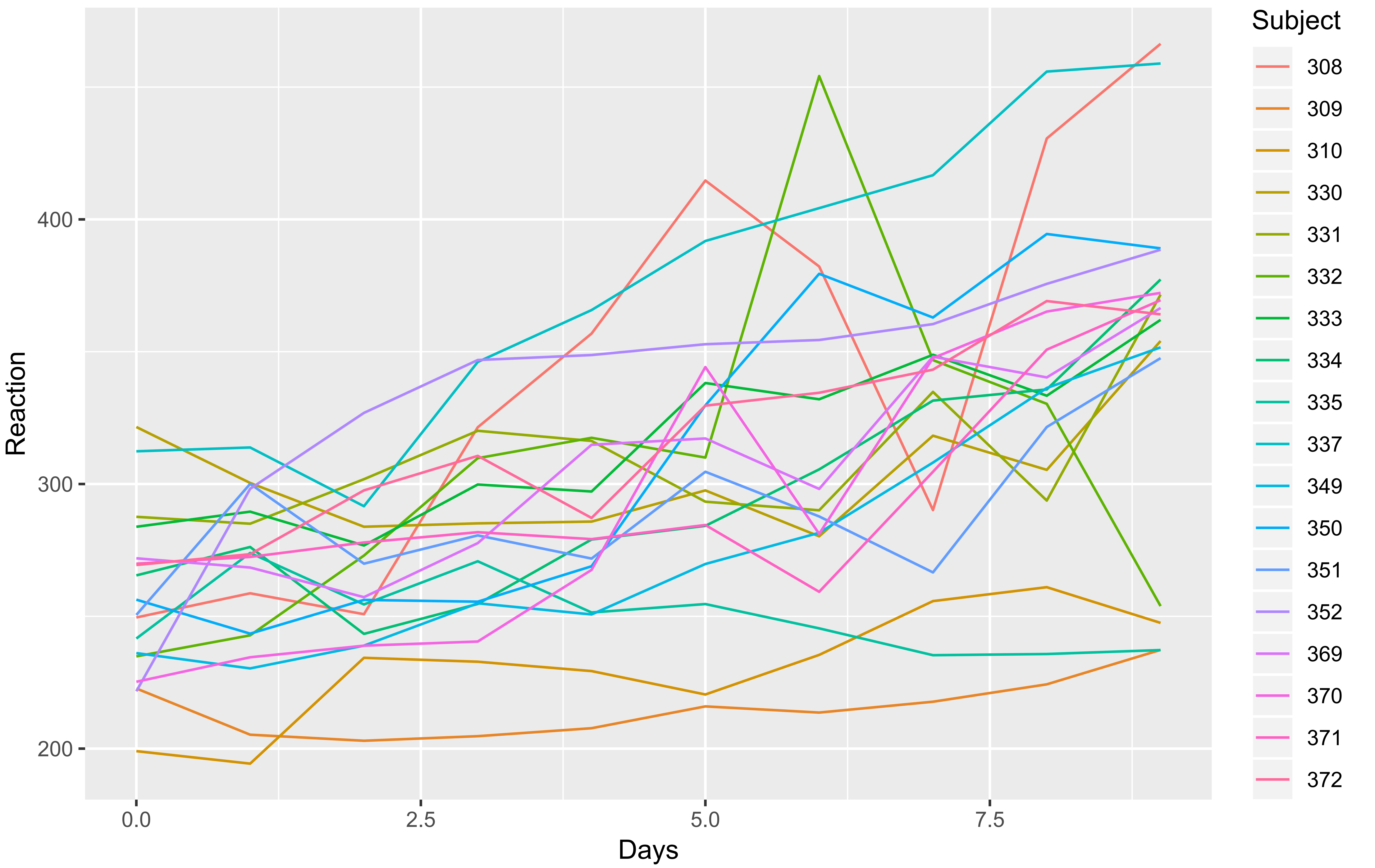
$$Y_{ti} = \beta_{0i} + \epsilon_{ti}$$
$$\beta_{0i} = \gamma_{00} + u_{0i}$$
$$u_{0i} \sim N(0,\ \tau_0^2)$$
$$\epsilon_{ti} \sim N(0,\ \sigma^2)$$

where Y_{ti} is the outcome for measurement t for person i

A SLEEP STUDY

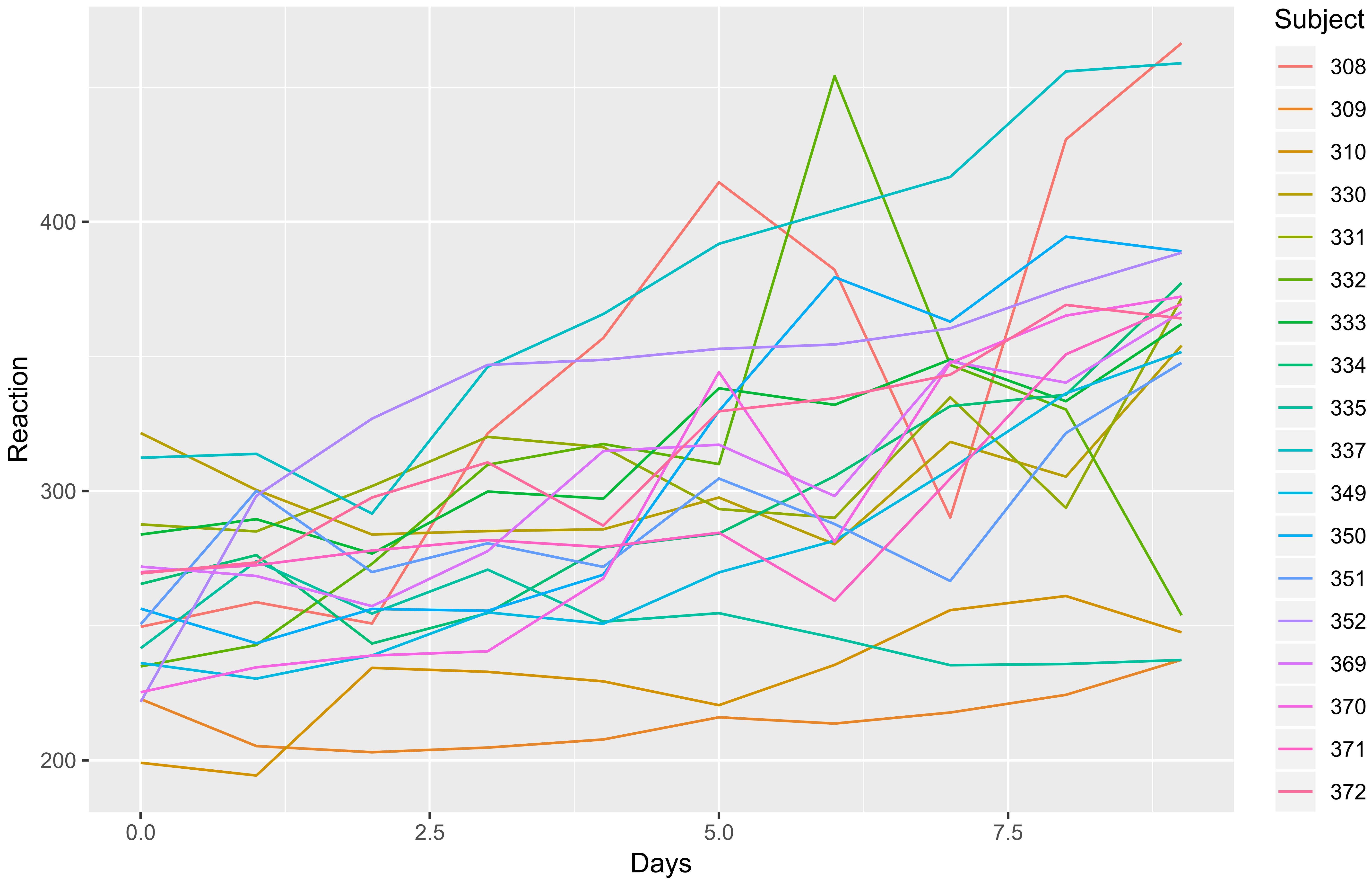
These data are from Belenky et al. (2003) in which reaction time was measured for participants after days of sleep deprivation (3 hrs per night).

18 subjects were measured 10 days and day 0 represents a normal night of sleep.



A SLEEP STUDY NULL MODEL

$$Y_{ij} = \beta_{0j} + \epsilon_{ij}$$
$$\beta_{0j} = \gamma_{00} + u_{0j}$$
$$u_{0j} \sim N(0,\ \tau_0^2)$$
$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$



These data are from Belenky et al. (2003) in which reaction time was measured for participants after days of sleep deprivation (3 hrs per night).

18 subjects were measured 10 days and day 0 represents a normal night of sleep.

AN UNCONDITIONAL GROWTH MODEL

$$Y_{ij} = \beta_{0j} + \beta_{1j}TIME_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right]$$

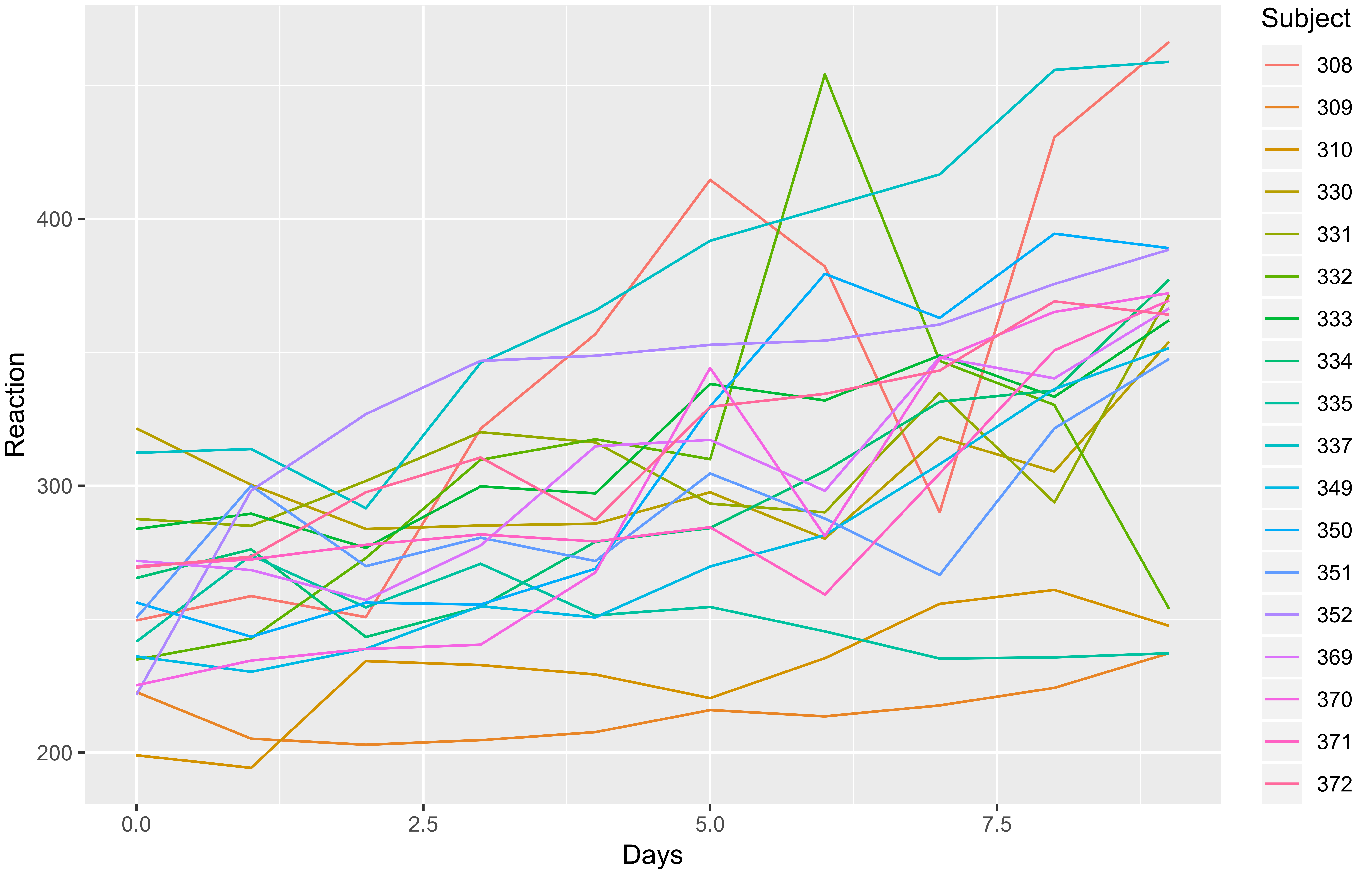
$$\epsilon_{ij} \sim N(0,\, \sigma^2)$$

MLMs with longitudinal data are often called growth models! Each person has their own line in the model!

Note that folks use the term "unconditional" here even though we have a covariate in the model.

It's unconditional in that we do not include any variables about the PEOPLE in the model!

A SLEEP STUDY GROWTH MODEL



R CODE FOR SLEEPSTUDY MODELS

```
fit = lmer(Reaction~1+ (1 | Subject), data=sleepstudy)

fit = lmer(Reaction ~ Days + (Days | Subject), data=sleepstudy)
```

A SLEEP STUDY: UNCONDITIONAL GROWTH MODEL

```
Random effects:
 Groups   Name        Variance Std.Dev. Corr
 Subject  (Intercept) 612.09   24.740
          Days         35.07    5.922   0.07
 Residual             654.94   25.592
Number of obs: 180, groups:  Subject, 18

Fixed effects:
            Estimate Std. Error t value
(Intercept)  251.405      6.825  36.838
Days          10.467      1.546   6.771
```

A CONDITIONAL GROWTH MODEL

$$Y_{ij} = \beta_{0j} + \beta_{1j} TIME_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01} Z_j + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + \gamma_{11} Z_j + u_{1j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix} \right)$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

We see that individuals have their own growth rates, but we are now curious if subject covariates predict some of those differences.

Growth modeling focuses primarily on adding in level 2 covariates (because we are usually interested in people).

But it is possible to include level 1 covariates about the measurements.

A CONDITIONAL GROWTH MODEL

$$Y_{ij} = \beta_{0j} + \beta_{1j} TIME_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{01} Z_j + u_{0j}$$

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$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right)$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

```
fit = lmer(Reaction ~
Days * Man +(Days |
Subject),
data=sleepstudy)
```

NOT A GROWTH MODEL, STILL A MLM

$$Y_{ij} = \beta_{0j} + \beta_1 TIME_{1ij} + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + \gamma_{10} Z_j + u_{0j}$$

$$\beta_1 = \gamma_{10}$$

$$u_{0j} \sim N(0, \tau_0^2)$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

```
fit = lmer(Reaction ~
Days + Man +(1 |
Subject),
data=sleepstudy)
```

A CONDITIONAL GROWTH MODEL

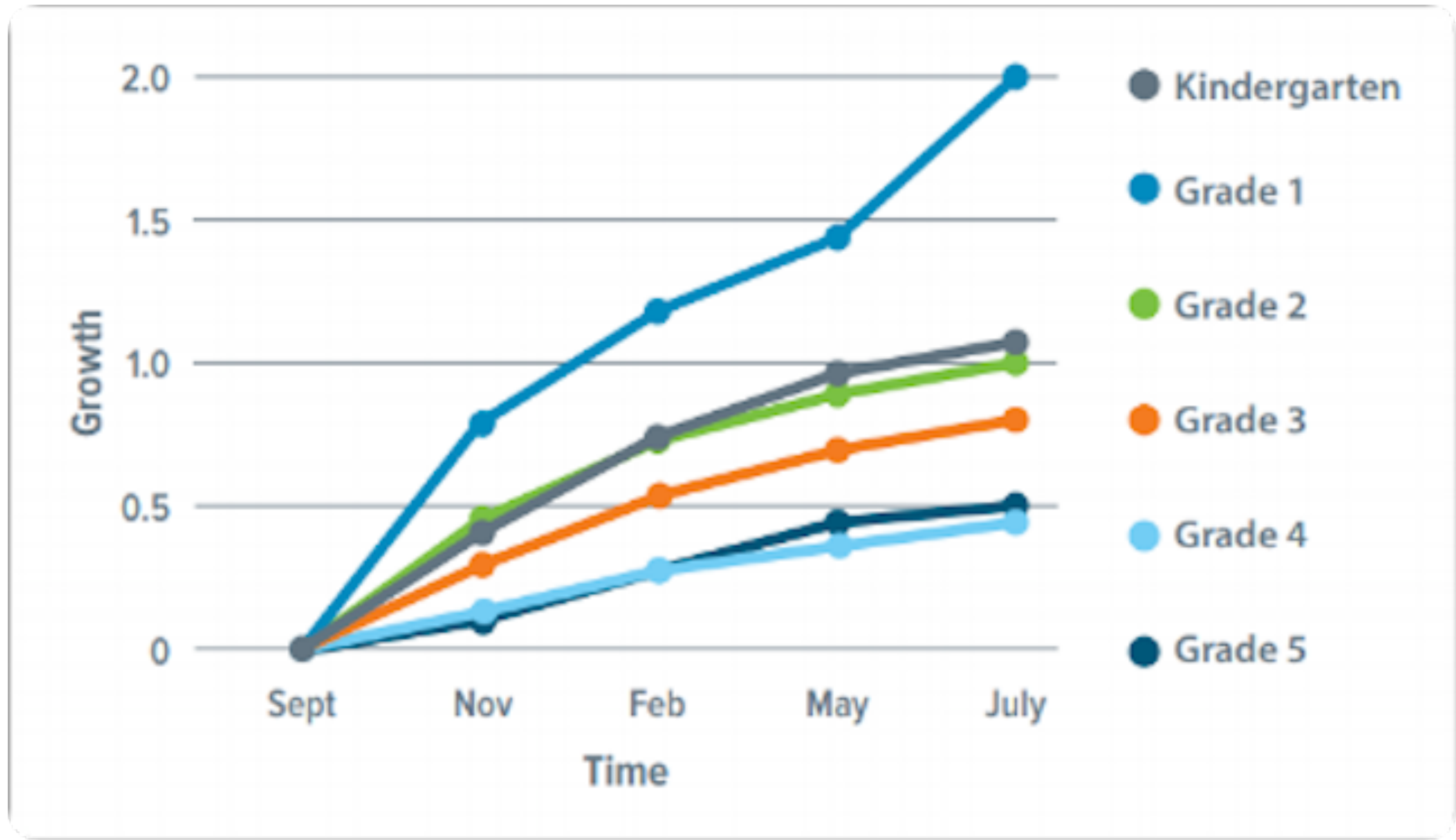
```
Random effects:
 Groups   Name        Variance Std.Dev. Corr
 Subject  (Intercept) 318.61   17.850
          Days         37.76    6.145   0.07
 Residual             654.94   25.592
Number of obs: 180, groups:  Subject, 18

Fixed effects:
             Estimate Std. Error t value
(Intercept) 263.80366    6.73833  39.150
Days         10.47693    1.95146   5.369
Man         -37.19565   11.67113  -3.187
Days:Man     -0.02893    3.38003  -0.009
```

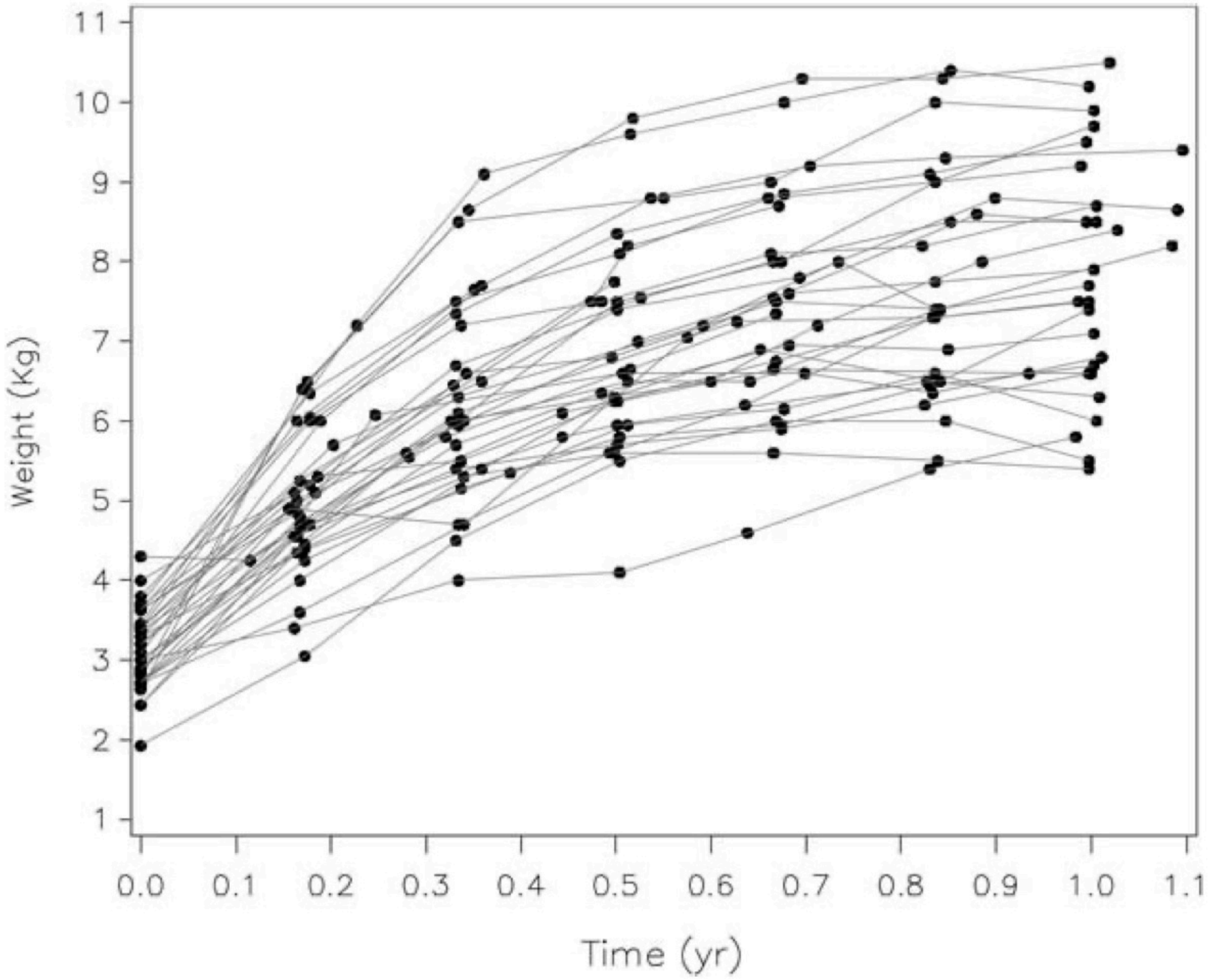
WHY A NONLINEAR MODEL?

Growth is often non-linear!

For some outcomes, we see folks increase at a steady rate at first and then begin to taper off.



From Dreambox website

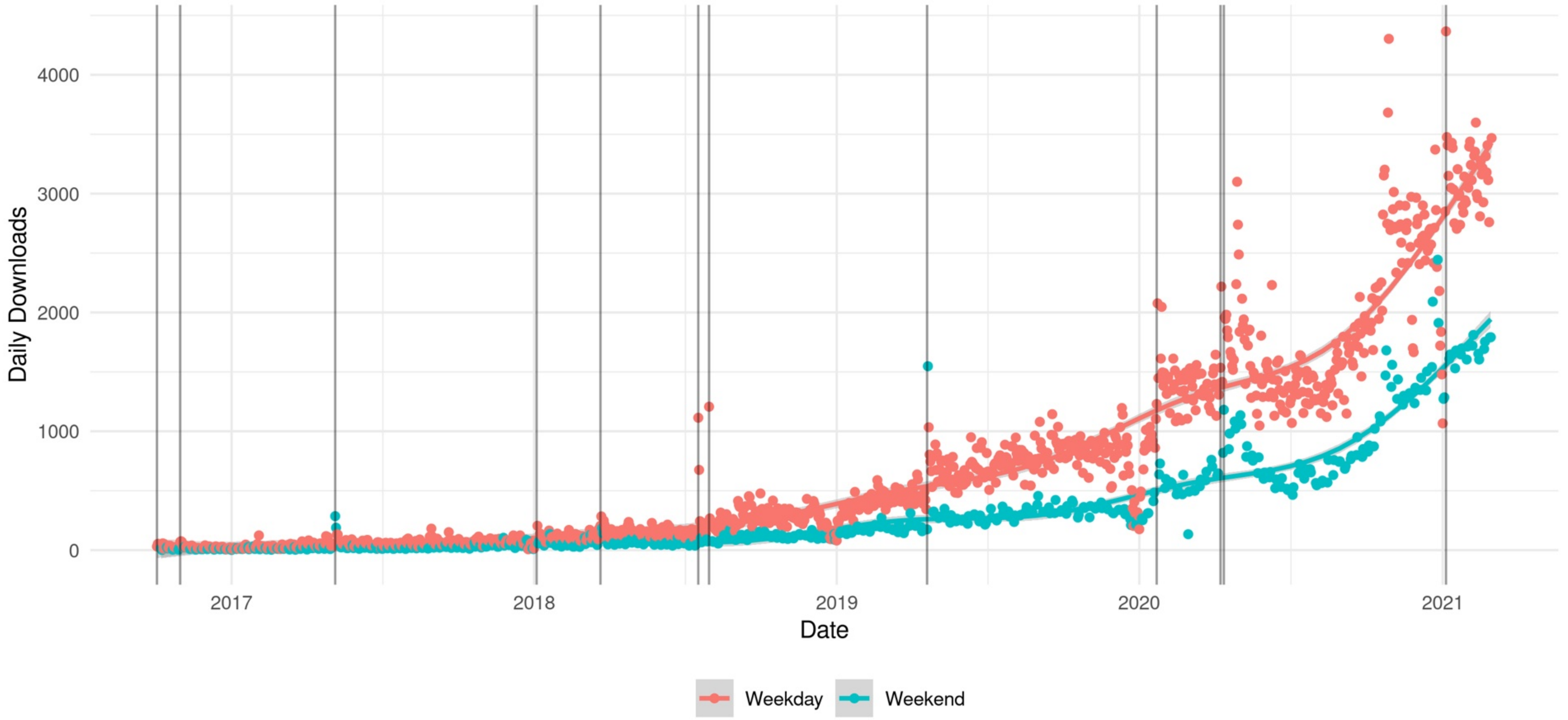


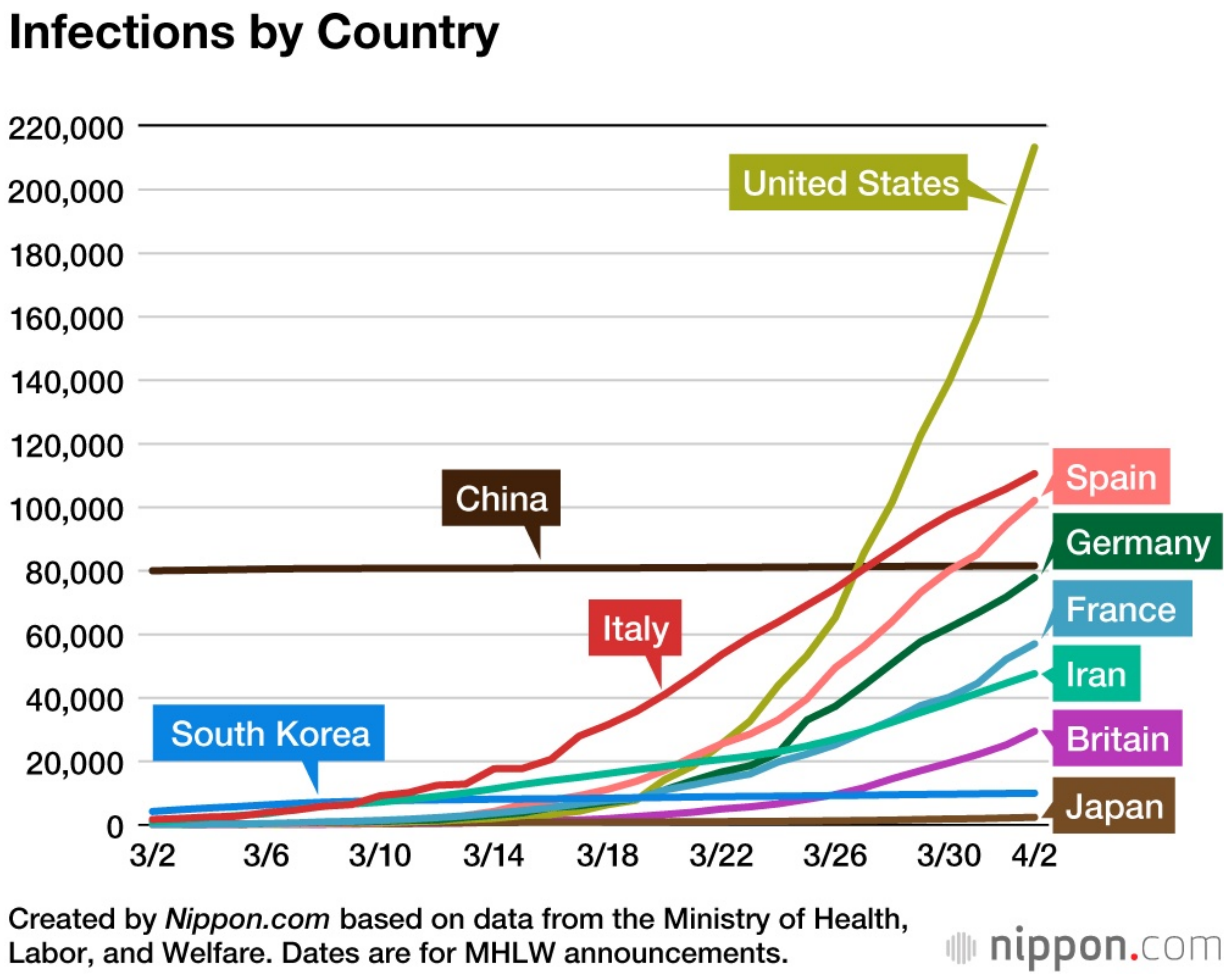
From Sterna (2014) Fitting Nonlinear Latent Growth Curve Models With Individually Varying Time Points

WHY A NONLINEAR MODEL?

Growth is often non-linear!

For some outcomes, we see folks increase at a steady rate at first and then begin to taper off.





HOW TO THINK ABOUT A QUADRATIC TERM

$$Y_{ij} = \beta_{0j} + \beta_{1j}TIME_{ij} + \beta_2 TIME_{ij}^2 + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\beta_{2j} = \gamma_{20}$$

A positive parameter (γ_{20}) for the quadratic term indicates a U shaped curve;

A negative parameter (γ_{20}) indicates an upside down U curve.

But we don't know which part of the curve we're on until we plot some data.

NONLINEAR UNCONDITIONAL GROWTH MODELS

$$Y_{ij} = \beta_{0j} + \beta_{1j}TIME_{ij} + \beta_{2j}TIME_{ij}^2 + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\beta_{2j} = \gamma_{20} + u_{2j}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \\ u_{2j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} {\tau_0}^2 & \tau_{01} & \tau_{02} \\ \tau_{10} & {\tau_1}^2 & \tau_{12} \\ \tau_{20} & \tau_{21} & \tau_2^2 \end{pmatrix} \right)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

$$Y_{ij} = \beta_{0j} + \beta_{1j}TIME_{ij} + \beta_2 TIME_{ij}^2 + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\beta_2 = \gamma_{20}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix} \right)$$

$$\epsilon_{ij} \sim N(0,\ \sigma^2)$$

R CODE FOR NONLINEAR UNCONDITIONAL GROWTH MODELS

```
lmer(Y ~ Time + I(Time^2) + (Time |Subject), data=MYDATA)

lmer(Reaction ~ Days + I(Days^2) + (Days |Subject), data=sleepstudy)
```

```
lmer(Y ~ Time + I(Time^2) + (Time+I(Time^2) |Subject), data=MYDATA)

lmer(Reaction ~ Days + I(Days^2) + (Days+ I(Days^2) |Subject),
data=sleepstudy)
```

INTERPRETING A FIXED QUADRATIC TERM

$$Y_{ij} = \beta_{0j} + \beta_{1j}TIME_{ij} + \beta_2 TIME_{ij}^2 + \epsilon_{ij}$$

$$\beta_{0j} = \gamma_{00} + u_{0j}$$

$$\beta_{1j} = \gamma_{10} + u_{1j}$$

$$\beta_2 = \gamma_{20}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right)$$

$$\epsilon_{ij} \sim N(0, \sigma^2)$$

```
Random effects:
 Groups   Name        Variance Std.Dev. Corr
 Subject  (Intercept) 613.12   24.761
          Days         35.11    5.925   0.06
 Residual             651.97   25.534
Number of obs: 180, groups:  Subject, 18

Fixed effects:
            Estimate Std. Error t value
(Intercept) 255.4494     7.5135  33.999
Days          7.4341     2.8189   2.637
I(Days^2)     0.3370     0.2619   1.287
```

3 LEVEL MULTILEVEL MODELS

3 LEVEL MODELS - SO MUCH NOTATION!

$$Y_{ijk} = \beta_{0jk} + \beta_{1jk}X_{ijk} + \epsilon_{ijk}$$

$$\beta_{0jk} = \gamma_{00k} + u_{0jk}$$

$$\beta_{1jk} = \gamma_{10k} + u_{1jk}$$

$$\gamma_{00k} = \delta_{000} + v_{0k}$$

$$\gamma_{10k} = \delta_{100} + v_{1k}$$

$$\epsilon_{ijk} \sim N(0,\sigma^2)$$

$$\begin{pmatrix} u_{0jk} \\ u_{1jk} \end{pmatrix} \sim MVN\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right)$$

$$\begin{pmatrix} v_{0k} \\ v_{1k} \end{pmatrix} \sim MVN\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \phi_0^2 & \phi_{01} \\ \phi_{10} & \phi_1^2 \end{pmatrix}\right)$$

3 LEVEL MODELS - SO MANY OPTIONS!

$$Y_{ijk} = \beta_{0jk} + \beta_{1k}X_{1ijk} + \beta_2 X_{2ijk} + \epsilon_{ijk}$$

$$\epsilon_{ijk} \sim N(0,\ \sigma^2)$$

$$\beta_{0jk} = \gamma_{00k} + u_{0jk}$$

$$\beta_{1k} = \gamma_{10k}$$

$$\beta_2 = \gamma_{20}$$

$$u_{0j} \sim N(0,\ \tau_0^2)$$

$$\gamma_{00k} = \delta_{000} + v_{0k}$$

$$\gamma_{10k} = \delta_{100} + v_{1k}$$

$$\begin{pmatrix} v_{0k} \\ v_{1k} \end{pmatrix} \sim MVN\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \phi_0^2 & \phi_{01} \\ \phi_{10} & \phi_1^2 \end{pmatrix}\right)$$

THE ONLY RULE IS...

$$Y_{ijk} = \beta_{0jk} + \beta_{1jk}X_{1ijk} + \epsilon_{ijk}$$

$$\beta_{0jk} = \gamma_{00k} + u_{0jk}$$
$$\beta_{1jk} = \gamma_{10k} + u_{1jk}$$

$$\gamma_{00k} = \delta_{000} + v_{0k}$$
$$\gamma_{10k} = \delta_{100} + v_{1k}$$

If something varies across level 2,

it will also vary by level 3.

3 LEVEL MODELS - RANDOM INTERCEPTS

Fitting 3-level multilevel models in lme4 is actually pretty simple for a random intercept only model. You can use either notation and you'll get the same thing.

```
fit = lmer(Y~X+(1|L3/L2), data=mydata)
fit = lmer(Y~X+(1|L3)+ (1|L3:L2), data=mydata)
```

$$Y_{ijk} = \beta_{0jk} + \beta_1 X_{ijk} + \epsilon_{ijk}$$

$$\beta_{0jk} = \gamma_{00k} + u_{0jk}$$

$$\beta_1 = \gamma_{10}$$

$$\gamma_{00k} = \delta_{000} + v_{0k}$$

$$\epsilon_{ijk} \sim N(0, \sigma^2)$$

$$u_{0j} \sim N(0, \tau_0^2)$$

$$v_{0j} \sim N(0, \phi_0^2)$$

3 LEVEL MODELS - RANDOM SLOPES AT LEVEL 2

Suppose we have students nested within schools nested within school districts. We may believe that association between homework and test scores varies by schools, which implies is also varies by districts.

```
fit = lmer(math~hw+(hw|L3/L2),
data=mydata)
```

```
fit = lmer(math~hw+(hw|L3)+
(hw|L3:L2), data=mydata)
```

$$Y_{ijk} = \beta_{0jk} + \beta_{1jk}X_{ijk} + \epsilon_{ijk}$$

$$\beta_{0jk} = \gamma_{00k} + u_{0jk}$$

$$\beta_{1jk} = \gamma_{10k} + u_{1jk}$$

$$\gamma_{00k} = \delta_{000} + v_{0k}$$

$$\gamma_{10k} = \delta_{100} + v_{1k}$$

$$\epsilon_{ijk} \sim N(0,\sigma^2)$$

$$\begin{pmatrix} u_{0jk} \\ u_{1jk} \end{pmatrix} \sim MVN\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right)$$

$$\begin{pmatrix} v_{0k} \\ v_{1k} \end{pmatrix} \sim MVN\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \phi_0^2 & \phi_{01} \\ \phi_{10} & \phi_1^2 \end{pmatrix}\right)$$

3 LEVEL MODELS - RANDOM SLOPES AT LEVEL 3

Suppose we have students nested within schools nested within school districts and that hwk is a fixed effect. We may believe that the effect of being a private school varies across districts.

```
fit = lmer(math~hw+private+
(private|L3)+ (1|L3:L2),
data=mydata)
```

Or we may also believe the effect of hwk is the same for all schools within in a district but varies by district.

```
fit = lmer(math~hw+(hw|L3)+
(1|L3:L2), data=mydata)
```

$$Y_{ijk} = \beta_{0jk} + \beta_1 X_{ijk} + \epsilon_{ijk}$$

$$\beta_{0jk} = \gamma_{00k} + \gamma_{01k} Z_{jk} + u_{0jk}$$

$$\beta_1 = \gamma_{10}$$

$$\gamma_{00k} = \delta_{000} + v_{0k}$$

$$\gamma_{01k} = \delta_{010} + v_{1k}$$

$$\epsilon_{ijk} \sim N(0,\, \sigma^2)$$

$$u_{0j} \sim N(0,\, \tau_0^2)$$

$$\begin{pmatrix} v_{0k} \\ v_{1k} \end{pmatrix} \sim MVN\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \phi_0^2 & \phi_{01} \\ \phi_{10} & \phi_1^2 \end{pmatrix} \right)$$

3 LEVEL MODELS - SO MANY INTERACTIONS!!

Suppose we have students nested within schools nested within school districts and that hwk varies by school. Suppose we think being a private school impacts this association.

```
fit = lmer(math~hw*private+(hw|
L3)+ (hw|L3:L2), data=mydata)
```

```
fit = lmer(math~hw*private+
(private|L3)+ (1|L3:L2),
data=mydata)
```

$$Y_{ijk} = \beta_{0jk} + \beta_{1jk}X_{ijk} + \epsilon_{ijk}$$

$$\beta_{0jk} = \gamma_{00k} + \gamma_{01k}Z_{jk} + u_{0jk}$$

$$\beta_{1jk} = \gamma_{10k} + \gamma_{11k}Z_{jk} + u_{1jk}$$

$$\gamma_{00k} = \delta_{000} + v_{00k}$$

$$\gamma_{10k} = \delta_{100} + v_{10k}$$

$$\gamma_{01k} = \delta_{010} + v_{01k}$$

$$\gamma_{11k} = \delta_{110} + v_{11k}$$

$$\begin{pmatrix} u_{0j} \\ u_{1j} \end{pmatrix} \sim MVN\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{10} & \tau_1^2 \end{pmatrix}\right)$$

$$\epsilon_{ijk} \sim N(0, \sigma^2)$$

$$\begin{pmatrix} v_{00k} \\ v_{10k} \\ v_{01k} \\ v_{11k} \end{pmatrix} \sim MVN\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \phi_{00}^2 & \phi_{00,10} & \phi_{00,01} & \phi_{00,11} \\ \phi_{10,00} & \phi_{10}^2 & \phi_{10,01} & \phi_{10,11} \\ \phi_{01,00} & \phi_{01,10} & \phi_{01}^2 & \phi_{01,11} \\ \phi_{11,00} & \phi_{11,10} & \phi_{11,01} & \phi_{11}^2 \end{pmatrix}\right)$$

MISSING DATA

MISSING DATA

Missing data are categorized based on our assumption of the mechanisms of missingnes

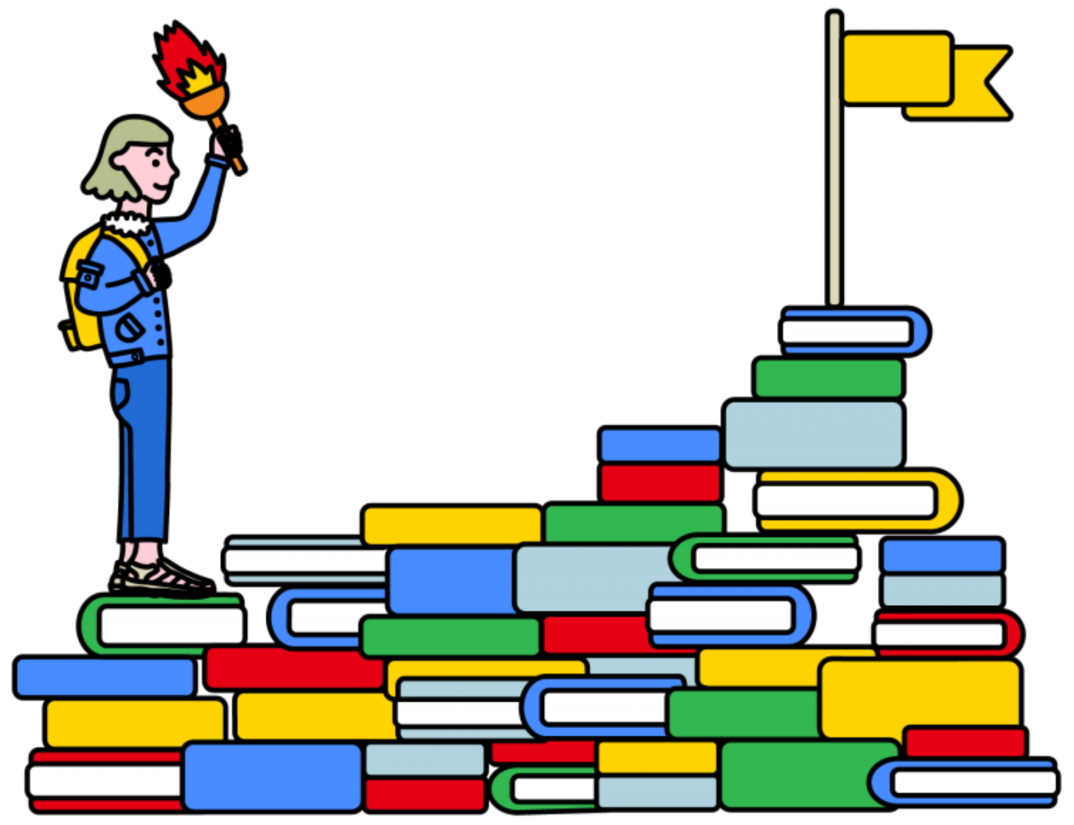
Let $\mathbf{Y} = (\mathbf{Y}_{obs}, \mathbf{Y}_{mis})$ and $\mathbf{R}$ be an indicator of missingness

- MCAR - Missing completely at random, $P(\mathbf{R} | \mathbf{Y}) = P(\mathbf{R})$, whether a particular observation is missing is independent of the data
- MAR - Missing at random, $P(\mathbf{R} | \mathbf{Y}) = P(\mathbf{R} | \mathbf{Y}_{obs})$, whether an observation is missing is independent of the unobserved (missing) data, given the observed data
- MNAR - Missing not at random, $P(\mathbf{R} | \mathbf{Y}) = P(\mathbf{R} | \mathbf{Y})$, whether an observation is missing is not independent of the missing data

MCAR, MAR, MNAR IN PRACTICE

We survey students about their motivation in math class. Assume we have data on who was sick and what club activities they attended.

1. Students who were sick do not take the survey. We assume that sickness does not impact motivation.
2. Students in the chess club were not in class to the take the survey, even though they tend to have higher motivation.
3. Students with low motivation skip those survey items.



MISSING DATA TREATMENT METHODS

Simple methods: Casewise deletion/complete case analysis; mean/mode imputation

Data-based methods: Hot deck imputation/nearest neighbor methods

Model-based methods: Regression imputation, Multiple imputation

Estimation-based methods: Full information maximum likelihood, Bayesian estimation

MISSING DATA TREATMENT METHODS

MCAR

Simple methods: Casewise deletion/complete case analysis; mean/mode imputation

Data-based methods: Hot deck imputation/nearest neighbor methods

Model-based methods: Regression imputation, Multiple imputation

Estimation-based methods: Full information maximum likelihood, Bayesian estimation

MAR

Model-based methods: Regression imputation, Multiple imputation

Estimation-based methods: Full information maximum likelihood, Bayesian estimation

MISSING DATA TREATMENT METHODS

MNAR

Model-based methods: Regression imputation, Multiple imputation

Estimation-based methods: Full information maximum likelihood, Bayesian estimation

$+$

Strong justifications for assumptions!

Sensitivity analyses!

MULTIPLE IMPUTATION

- Don Rubin introduced the concept of multiple imputation (MI) in the 1970s, and it's considered the gold standard for missing data
- It's based on Bayesian estimation, but MI caught on more quickly
- It's basically regression imputation that accounts for residual error (σ^2) in the predictions!

$$W_{obs} = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p + \epsilon$$

Rubin, D. B. (1996). Multiple Imputation after 18+ Years. Journal of the American Statistical Association, 91(434), 473–489. https://doi.org/10.1080/01621459.1996.10476908

WHAT IF MULTIPLE COVARIATES ARE MISSING?

There are generally two “good” approaches to multiple imputation:

Joint Modeling (JM)

A single model is used for imputation, where anything on the right side of the model is fully observed.

Your model could have multiple covariates on the left, hence the name joint modeling.

Implemented in the `pan` package

Fully Conditional Specification(FCS)

A separate model is used for each covariate with missingness.

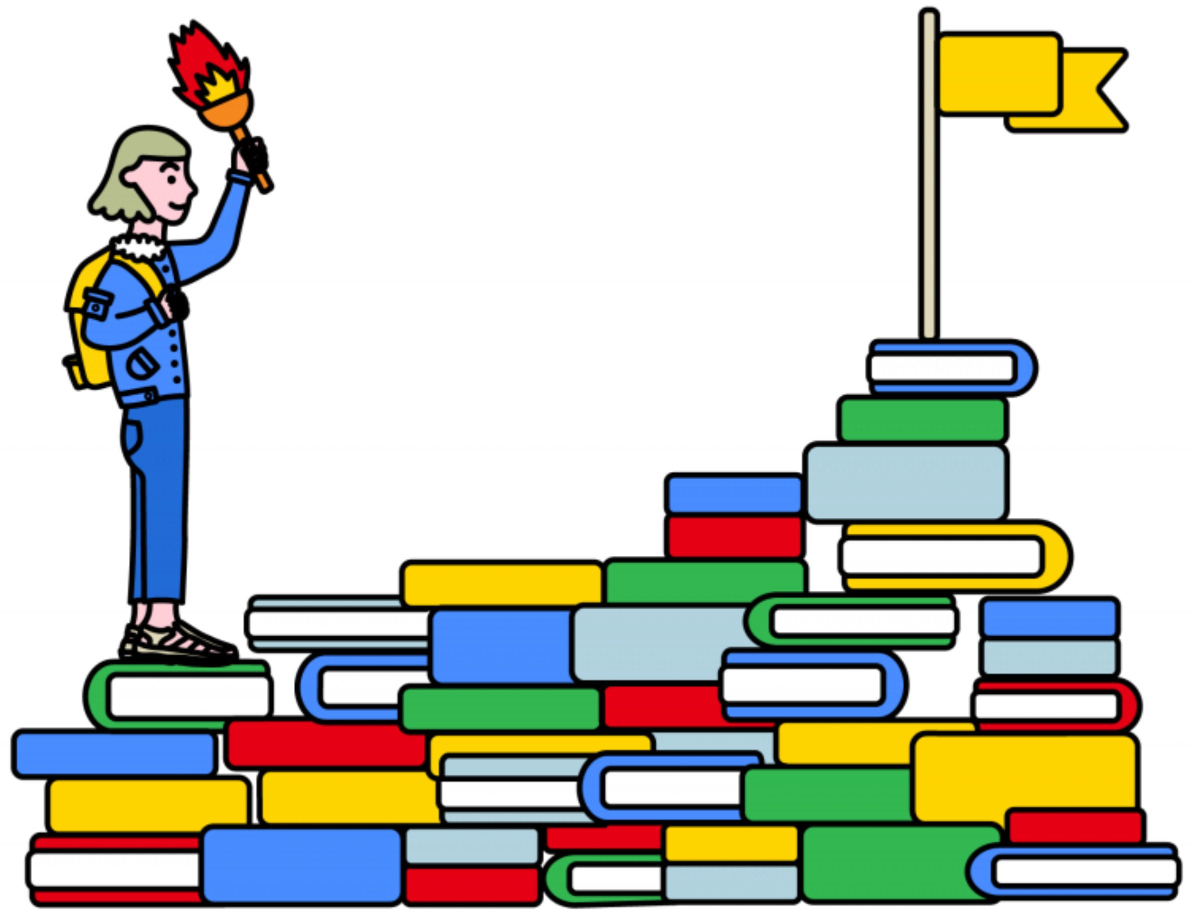
We iterate back and forth across the models continuing to impute missing data until some threshold is achieved.

Implemented in the `mice` package

MCAR, MAR, MNAR IN PRACTICE

We survey students about their motivation in math class, using a multistage sampling design. Suppose that liking your teacher is related to your motivation in math class.

1. Several schools forgot to administer the survey.
2. In about half the schools, only the most well-liked teachers administered the survey.
3. Students with lower motivation tended not to answer the survey items.



4. Schools where liking your teacher is less associated with your motivation were less likely to administer the survey.
5. Private schools were more likely to administer the survey.

MITML R PACKAGE

This is an R package that allows users to impute missing data at level 1 or level 2 using the Joint Modeling method.

There is code in the Grund et al. paper for Multilevel FCS, but it's more complicated.

Just note that a random slope/interaction model may not work out well with MITML when data are MAR.

From Grund, Lüdtke &Robitzsch (2018) DOI: 10.1177/1094428117703686

Table 6. Recommended Missing Data Treatments and Software for Different Types of Multilevel Analysis Models Recommended Missing Data Treatments and Software for Different Types of Multilevel Analysis Models

Model type (Example)	Missing	Recommended	Not Recommended	Current Software (MI)
Random intercept model $Y_{ij} = \gamma_{00} + \gamma_{10}(X_{ij} - \bar{X}_{\bullet j}) + \gamma_{01}\bar{X}_{\bullet j} + u_{0j} + e_{ij}$	Any	• Multilevel FCS ▹ Passive imputation of group means • Multilevel JM ▹ All variables specified as targets	• Listwise deletion ▹ Biased estimates, power loss • Single-level MI ▹ biased estimates and SEs • FIML[a] ▹ Biased estimates when using group-mean centering	R (mice, pan, jomo), M*plus*, Blimp, SAS (MMI_IMPUTE), MLwiN, REALCOM → see Example 1.1
. . . with categorical variables (D_{ij}) $Y_{ij} = \gamma_{00} + \gamma_{10}D_{ij} + \gamma_{20}X_{ij} + u_{0j} + e_{ij}$	*D*	• Multilevel FCS ▹ Passive imputation of group means ▹ Using logistic or probit models • Multilevel JM ▹ All variables specified as targets ▹ Using models for mixed data types	• Listwise deletion ▹ Biased estimates, power loss • Single-level MI ▹ Biased estimates and SEs • FIML[a] ▹ Biased estimates under normality assumption	R (mice, jomo), M*plus*, Blimp, REALCOM → see Example 1.2
. . . with variables at Level 2 (W_j) $Y_{ij} = \gamma_{00} + \gamma_{10}(X_{ij} - \bar{X}_{\bullet j}) + \gamma_{01}\bar{X}_{\bullet j} + \gamma_{02}W_j + u_{0j} + e_{ij}$	*W*	• Multilevel FCS ▹ Including group means • Multilevel JM ▹ All variables specified as targets ▹ Using models for missing data at both levels • FIML	• Listwise deletion ▹ Biased estimates, power loss • Single-level MI ▹ Biased estimates and SEs	R (mice, jomo), M*plus*, Blimp, REALCOM → see Example 1.3
. . . with interactions or nonlinear terms $Y_{ij} = \gamma_{00} + \gamma_{10}X_{ij} + \gamma_{20}Z_{ij} + \gamma_{30}X_{ij}Z_{ij} + u_{0j} + e_{ij}$	*X*, *Z*	• Multilevel FCS ▹ Passive imputation of group means and product terms	• Listwise deletion ▹ Biased estimates, power loss • Single-level MI ▹ Biased estimates and SEs • Multilevel JM ▹ Biased estimates of interaction effects • FIML[a]	R (mice) → see Example 1.4
Random slope model $Y_{ij} = \gamma_{00} + \gamma_{10}(X_{ij} - \bar{X}_{\bullet j}) + \gamma_{01}\bar{X}_{\bullet j} + u_{0j} + u_{1j}(X_{ij} - \bar{X}_{\bullet j}) + e_{ij}$	Any	• Multilevel FCS ▹ Passive imputation of group means ▹ Including random slopes between pairs of variables	• Listwise deletion ▹ Biased estimates, power loss • Single-level MI ▹ Biased estimates and SEs • Multilevel JM ▹ Biased SEs • FIML[a]	R (mice), Blimp → see Example 2.1
. . . with interactions or nonlinear terms $Y_{ij} = \gamma_{00} + \gamma_{10}(X_{ij} - \bar{X}_{\bullet j}) + \gamma_{01}\bar{X}_{\bullet j} + \gamma_{02}W_j + \gamma_{11}W_j(X_{ij} - \bar{X}_{\bullet j}) + \gamma_{03}W_j\bar{X}_{\bullet j} + u_{0j} + u_{1j}(X_{ij} - \bar{X}_{\bullet j}) + e_{ij}$	*X*, *W*	• Multilevel FCS ▹ Passive imputation of group means and product terms ▹ Including random slopes between pairs of variables	• Listwise deletion ▹ Biased estimates, power loss • Single-level MI ▹ Biased estimates and SEs • Multilevel JM ▹ Biased estimates of interaction effects and SEs • FIML[a]	R (mice) → see Example 2.2

[a]The present recommendations refer to FIML as it is currently implemented in the statistical software M*plus*.